



UNIVERSITI
MALAYA
*Faculty of Computer Science
and Information Technology*

WOX7001 – RESEARCH METHODOLOGY

Topic 3 – Research Method in Computer Science

Agenda

01

SCIENTIFIC
METHOD

02

RESEARCH
CLASSIFIER

03

RESEARCH
METHODS

04

RESEARCH
PARADIGM

05

CLASS
ACTIVITIES

Scientific method

- Scientists use observations and reasoning to develop technologies and propose explanations for natural phenomena in the form of hypotheses
- Predictions from these hypotheses are tested by experiment and further technologies developed
- Any hypothesis which is cogent enough to make predictions can then be tested reproducibly in this way
- Once it has been established that a hypothesis is sound, it becomes a theory.

Elements of Scientific Method

- Characterisations (Quantifications, observations and measurements)
- Hypotheses (theoretical, hypothetical explanations of observations and measurements)
- Predictions (reasoning including logical deduction from hypotheses and theories)
- Experiments (tests of all of the above)

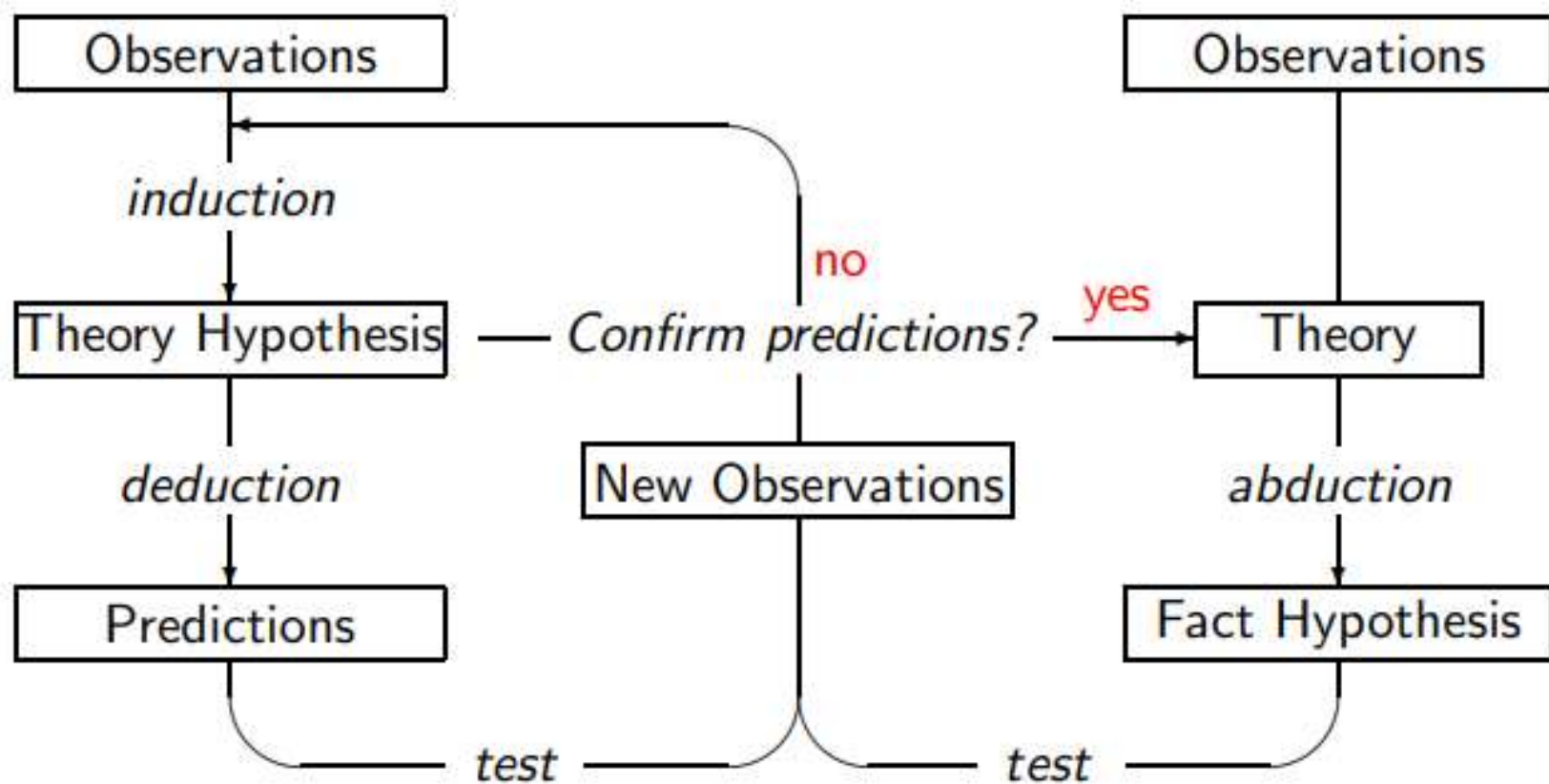
Scientific Method discovers....

- What predictions does a theory make?
- What is the right hypothesis in a particular situation?
- What is the right experiment to conduct?

The process involved in deriving the scientific method....

- Deductive reasoning (proceeds from our knowledge of the world (theories) and predicts 'likely' observations *Example: – Assume we know that A implies B. – A has been observed. – Then we should also observe B. – often lead to new knowledge*
- Abductive reasoning (proceeds from observations to causes. *Example: – The phenomenon X is observed. – Among hypotheses A, B, C, and D, only A and B are capable of explaining X. – Hence, there is a reason to assume that A or B holds. Useful for hypothesis generation*
- Inductive reasoning (proceeds from a set of observations to a general conclusion *Example: – Tycho Brahe, a 16th century astronomer, collected data on the movement of the Mars. Useful for theory formation*

The model for scientific method



Research Classifier

Prepared by: HSM, NAG and SUH

Research Perspective

Field (Position of the research within a hierarchy of topics)

Example:

Artificial Intelligence → Automated Reasoning →
First-Order Reasoning → Decidability

Approach (Research methods that are employed as part of the research process)

Examples:

Case study, Experiment, Survey, Proof

Nature (Pure theoretical development)

Review of pure theory and evaluation of its applicability

Applied research

Research Classification

Pure Theory	Descriptive studies	Exploratory Studies
Explanatory studies	Causal studies	Normative studies
Empirical Research	Problem solving studies	Development and application studies

Research Methods

Prepared by: HSM, NAG and SUH

Research Approach

Quantitative research methods Method

associated with measurements (on numeric scales) Stemming from natural sciences

Used to test hypotheses or create a set of observations for inductive reasoning

Accuracy and repeatability of vital importance

Qualitative research Methods

Methods involving case studies and surveys Stemming from social sciences

Concerned with increasing understanding of an are, rather than an explanation

Repeatability usually a problem

Research Methods

Action research:

Pursues action (or change) and understanding at the same time

Continuously alternates between action and critical reflection, while refining methods, data and interpretation in the light of the understanding developed in the earlier cycles

Example: Reflective teaching

Case study:

In-depth exploration of a single situation

Usually generates a large amount of (subjective) data

Should not merely report the data obtained or behaviour observed but attempt to generalise from the specific details of the situation observed

Example: Case study of open source software development

Research Methods

Survey:

Usually undertaken using questionnaires or interviews

Questionnaire and interview design important!

Determination of sample size and sample elements important!

Example: Survey on the popularity or use of programming languages

Experiment:

Investigation of causal relationships using test controlled by the researcher

Usually performed in development, evaluation and problem solving projects

Example: Evaluation of processor performance

Research Paradigm in CS

Prepared by: HSM, NAG and SUH

Research Paradigm in CS

Empirical:

Computer science is concerned with the study of a class of phenomena

Mathematical:

Computer Science is concerned with the study of algorithms and properties of information structures (abstraction from real objects)

Engineering:

managing the cost-effective design and construction of complex software-hardware systems (commercially and socially valuable)

Experimental Techniques

Depending on the objective, various evaluation techniques shall be used

Quantitative testing/experiments of algorithms/programs/databases/...

Usability tests with users

Questionnaires

Surveys

Case studies

Parameters.....

Runtime

Preprocessing time

Disk space (overhead)

Memory

Correctness of results

Accuracy of approximation algorithms

User satisfaction

Usability

So, in order to experiment we need Data set

Real-world data

- Always good to have – show that system works in practice
- Sometimes difficult to obtain
- Do not allow to test all aspects of an algorithm/system

Synthetic data

- Allow to test specific aspects of the algorithm
- Often (very) difficult to generate
- If possible, try to use the same data as your competitors

It is easy to show that your approach is better if only very particular data is used
Describe the most important aspects of the data

After experiment, compare the solution (Benchmarks)

Use existing benchmarks as much as possible

Facilitates the comparison of different solutions

Class Activity

Estimated time – 1 hour

Identify the following for your research title:

1. Which scientific method is suitable?
2. It falls under which research perspective?
3. The research classification?
4. Research approach?
5. Research method?
6. It belongs to which research paradigm in CS?
7. What technique to align with the selected research paradigm?

THANK YOU

Prepared by: HSM, NAG and SUH



WOX7001-RESEARCH METHODOLOGY

Topic 5 - Quantitative Research

Agenda

01

SAMPLE

02

RESEARCH
ETHICS

03

RESEARCH
DATA

04

RESEARCH
INSTRUMENT

05

CLASS
ACTIVITIES



Quantitative Research is...

- Quantitative research is the process of collecting and analyzing numerical data
- is a systematic investigation of phenomena by gathering quantifiable data and performing statistical, mathematical, or computational techniques
- collects information from existing and potential customers using sampling methods and sending out online surveys, online polls, and questionnaires
- can be used to find patterns and averages, make predictions, test causal relationships, and generalize results to wider populations.
- widely used in the natural and social sciences: biology, chemistry, psychology, economics, sociology, marketing, etc.



Characteristics of quantitative research ...

Structured tools:

- Structured tools such as surveys, polls, or questionnaires are used to gather quantitative data.
- Using such structured methods helps in collecting in-depth and actionable data from the survey respondents.

Sample size:

- conducted on a significant sample size that represents the target market.
- Appropriate sampling methods have to be used when deriving the sample to fortify the research objective

Close-ended questions:

- Closed-ended questions are created per the objective of the research.
- help collect quantitative data and hence, are extensively used in quantitative research.



Characteristics of quantitative research ...

Prior studies:

- Various factors related to the research topic are studied before collecting feedback from respondents.

Quantitative data:

- Usually, quantitative data is represented by tables, charts, graphs, or any other non-numerical form.
- This makes it easy to understand the data that has been collected as well as prove the validity of the market research.

Generalization of results:

- Results of this research method can be generalized to an entire population to take appropriate actions for improvement.



Advantages of quantitative research...

- Collect reliable and accurate data
- Quick data collection
- Wider scope of data analysis
- Eliminate bias



Population and Sample

- A population is a group of individuals that comprise the same characteristics
- A sample is a sub-group of the target population that the researcher plans to study for the purpose of making generalizations about the target population.
- Samples are only estimates
- The difference between the sample estimate and the true population is the “sampling error.”



Who Will You Study?

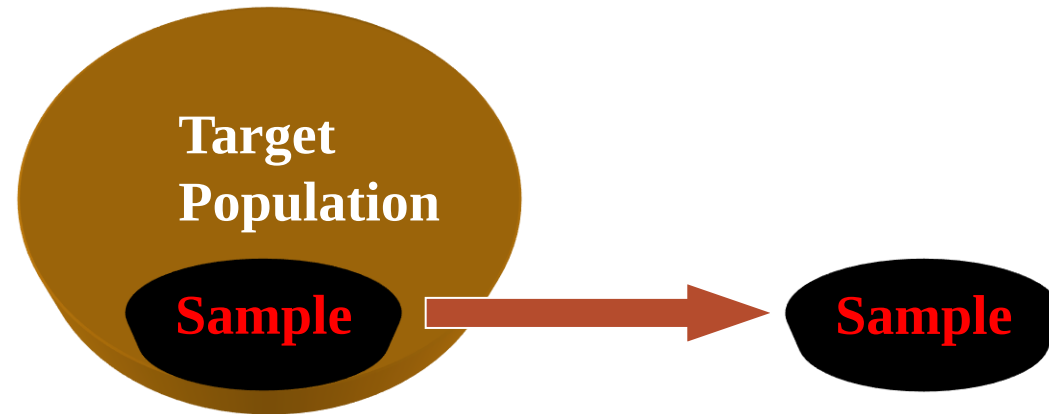
Identifying The Unit of Analysis

- Unit of analysis is the level (e.g. individual, family, school, school district) at which the data will be gathered.
- There may be different units of analysis
- one for the dependent variable
- one for the independent variable



Population and Samples

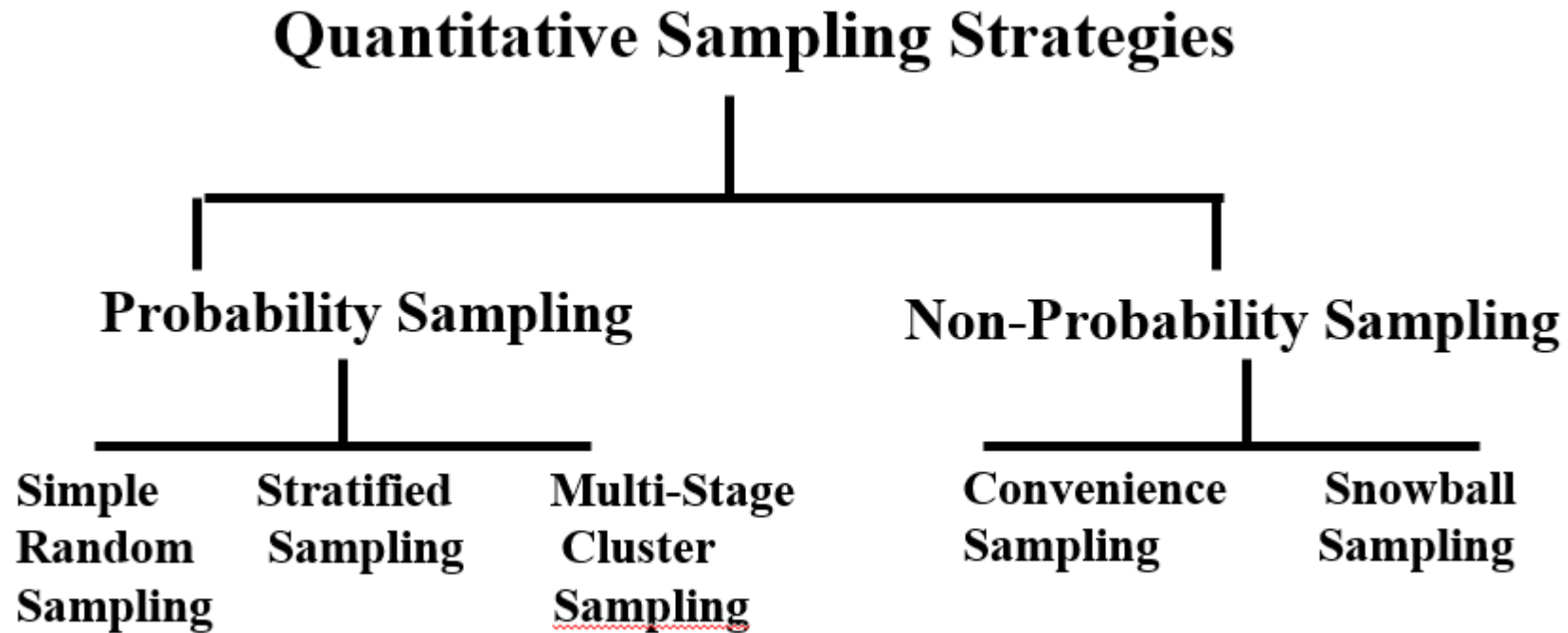
All teachers in high schools in one city
College students in all community colleges
Adult educators in all schools of education



All high school biology teachers
Students in one community college
Adult educators in 5 schools of education in the Midwest



Type of Quantitative Sampling





Types of Probability Samples

Simple Random:

selecting a sample from the population so all in the population have an equal chance of being selected

Systematic:

choosing every “nth” individual or site in the population until the desired sample size is achieved



Types of Probability Samples

Multi-Stage Cluster Sampling:

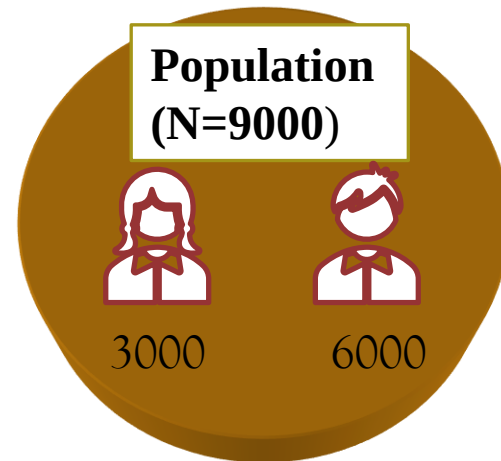
a sample chosen in one or two stages because the population is not easily identified or is large

Stratified sampling:

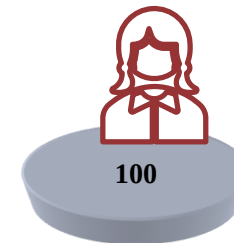
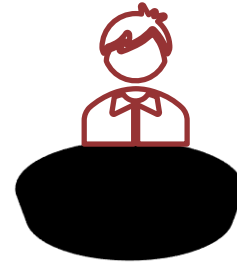
stratifying the population on a characteristic (e.g. gender) then sampling from each stratum.



Proportional Stratification Sampling Approach



Let's say from literature I have decided go for 3% from population



Sample = 300



Types of Non-Probability Samples

Convenience Sampling:

participants are selected because they are willing and available to be studied

Snowball Sampling:

the researcher asks participants to identify other participants to become members of the sample.

Research Ethics Application

Prepared by: HSM, NAG and SUH



What Permissions Will You Need? Obtaining Permission

Institutional or organizational (e.g. school district)

Site-specific (e.g. secondary school)

Individual participants or parents

Campus approval (e.g. university or college) and Institutional Review Board (IRB)

Research Data and Instrument

Prepared by: HSM, NAG and SUH



Research Instrument

An instrument is a tool for measuring, observing, or documenting quantitative data

Types of Instruments

Performance Measures (e.g. test performance)

Attitudinal Measures (measures feelings toward educational topics)

Behavioral Measures (observations of behavior)

Factual Measures (documents, records)



Linking data collection to variables and questions

Identify the variable



Operationally define the variable



Locate data (measures, observations, documents with questions and scales)



Collect data on instruments yielding numeric scores

Self-efficacy for learning from others



Level of confidence that an individual can learn something by being taught by others



13 items on a self-efficacy attitudinal scale from Bergin (1989)



Scores of each item ranged from 0-10 with 10 being "completely confident."



What Instruments Will You Use To Collect Data? Locating or Developing an Instrument

Look in published journal articles

Run an ERIC search and use the term “instruments” and the topic of the study

Go to ERIC web site for Evaluation and Assessment

Examine guides to commercially available tests

Develop your own instrument



Criteria for choosing a good instrument:

Reliability

Reliability: Scores from measuring variables that are stable and consistent

Example: Bathroom scale

Types of reliability

- Test-retest (scores are stable over time)

- Alternate forms (equivalence of two instruments)

- Alternate forms and test-retest

- Inter-rater reliability (similarity in observation of a behavior by two or more individuals)

- internal consistency (consistent scores across the instrument)



Criteria for choosing a good instrument: Validity

Validity: Scores from measuring variables that are meaningful

Types of validity

- Content (representative of all possible questions that could be asked)

- Criterion-referenced (scores are a predictor of an outcome or criterion they are expected to predict)

- Construct (determination of the significance, meaning, purpose and use of the scores)



Criteria for choosing a good instrument: Scales of Measurement

Nominal (Categorical): categories that describe traits or characteristics participants can check

Ordinal (Categorical): participants rank order a characteristic, trait or attribute

Interval (Continuous): provides “continuous” response possibilities to questions with assumed equal distance

Ratio (Continuous): a scale with a true zero and equal distances among units



How Will You Administer the Data Collection? Procedures for Administering the Data Collection

Develop standard written procedures for administering an instrument

Train researchers to collect observational data

Obtain permission to collect and use public documents

Respect individuals and sites during data gathering (ethics)



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After experiment, compare the solution (Benchmarks)

Use existing benchmarks as much as possible

Facilitates the comparison of different solutions



Class Activity

Estimated time – 1 hour

Refer to the article given, answer the following questions

Structured Tool	
Population	
Sample	
Probability/Non Probability	
Types of sampling	
Prior studies	
Hypothesis	
Findings	

THANK YOU

Collecting Quantitative Data



Key Ideas

- Who will you study (unit, sampling, sample size)?
- What permissions will you need? (levels, IRB)
- What information will you collect? (types of data, links to questions/variables)
- What instrument(s) will you use? (selecting an instrument, scales of measurement, validity, reliability)
- How will you administer the data collection? (standardization, ethical issues)

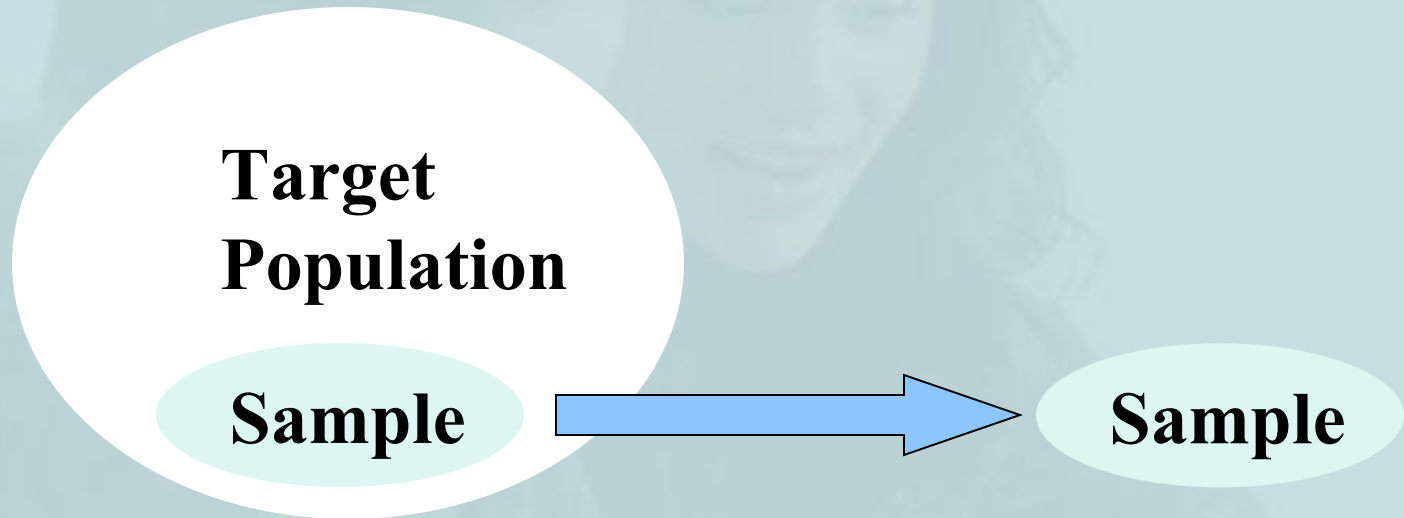
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Populations and Samples



Population

- All teachers in high schools in one city
- College students in all community colleges
- Adult educators in all schools of education

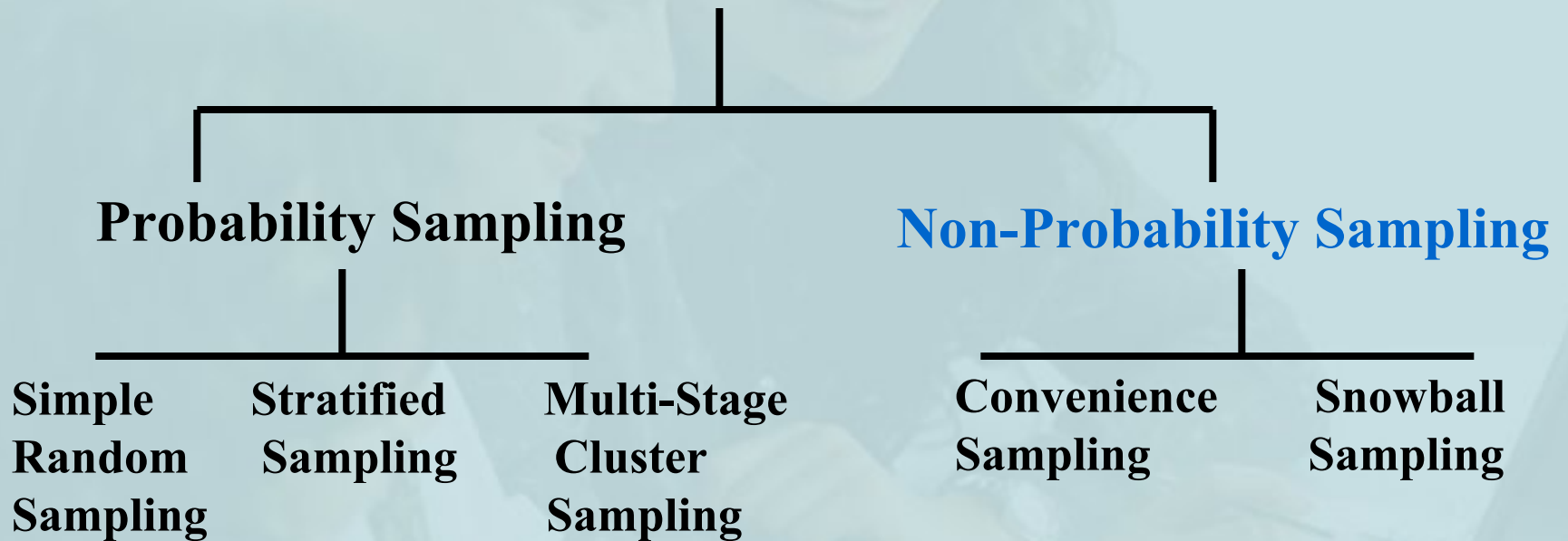
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Probability and Non-Probability Sampling

- **Probability sampling** is the selection of individuals from the population so that they are representative of the population
- **Non-probability** sampling is the selection of participants because they are available, convenient, or represent some characteristic the investigator wants to study.

Types of Quantitative Sampling

Quantitative Sampling Strategies



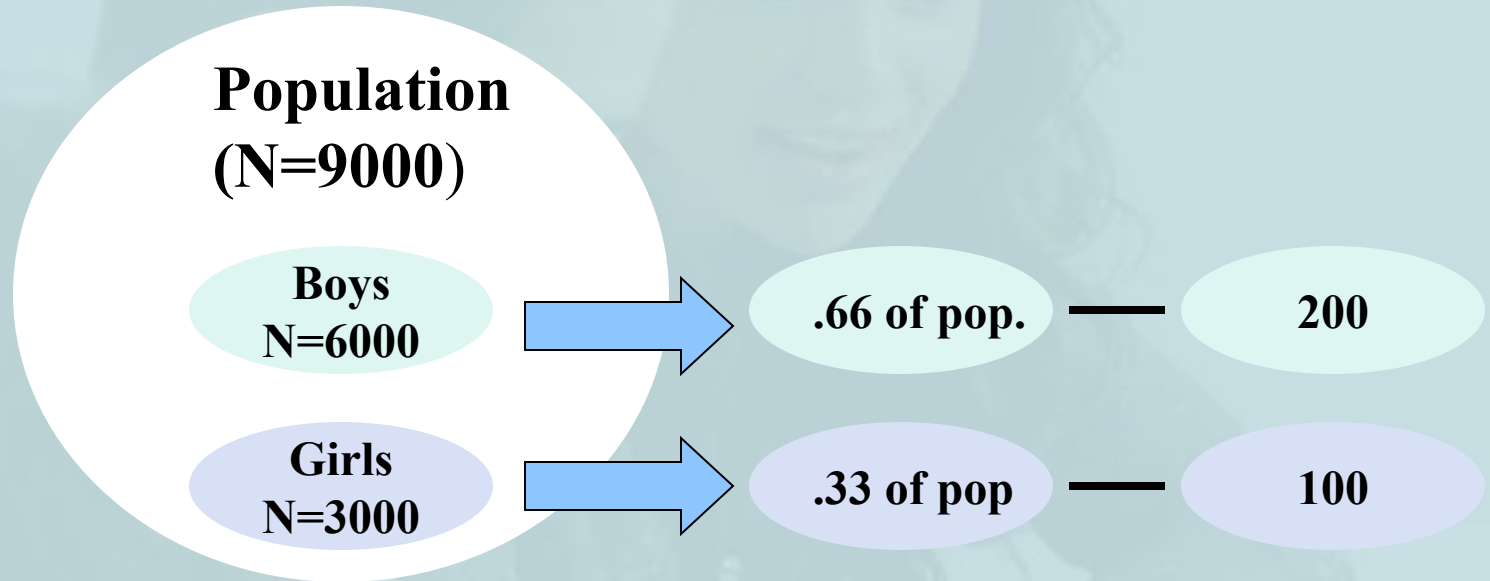
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What Information Will You Collect? Types of data measures

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- Types of Instruments
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Linking data collection to variables and questions

Flow of Activities

Identify the variable



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Example

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 - Construct (determination of the significance, meaning, purpose and use of the scores)

Construct Validity

Example

Self Esteem: A useful variable to have scores on for high school students, parents, and counselors

Contrary Example

Carrying knives by a street gang. Not meaningful information. Everyone in the gang does it.

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Analyzing and Interpreting Quantitative Data



Steps in quantitative analysis and interpretation

- How do you prepare the data for analysis?
- How do you analyze the data?
- How do you report the results?
- How do you discuss the results?

How do you prepare the data for analysis? Inputting data

- Score data by assigning numeric codes to responses
- Create codebook
- Use information from instrument when possible as part of coding scheme
- Create data file in data grid
- Create variable, value labels
- Clean database, missing values

Determine types of Scores to analyze

- Single item
- Summed scores
- Difference scores

Selecting a statistical program

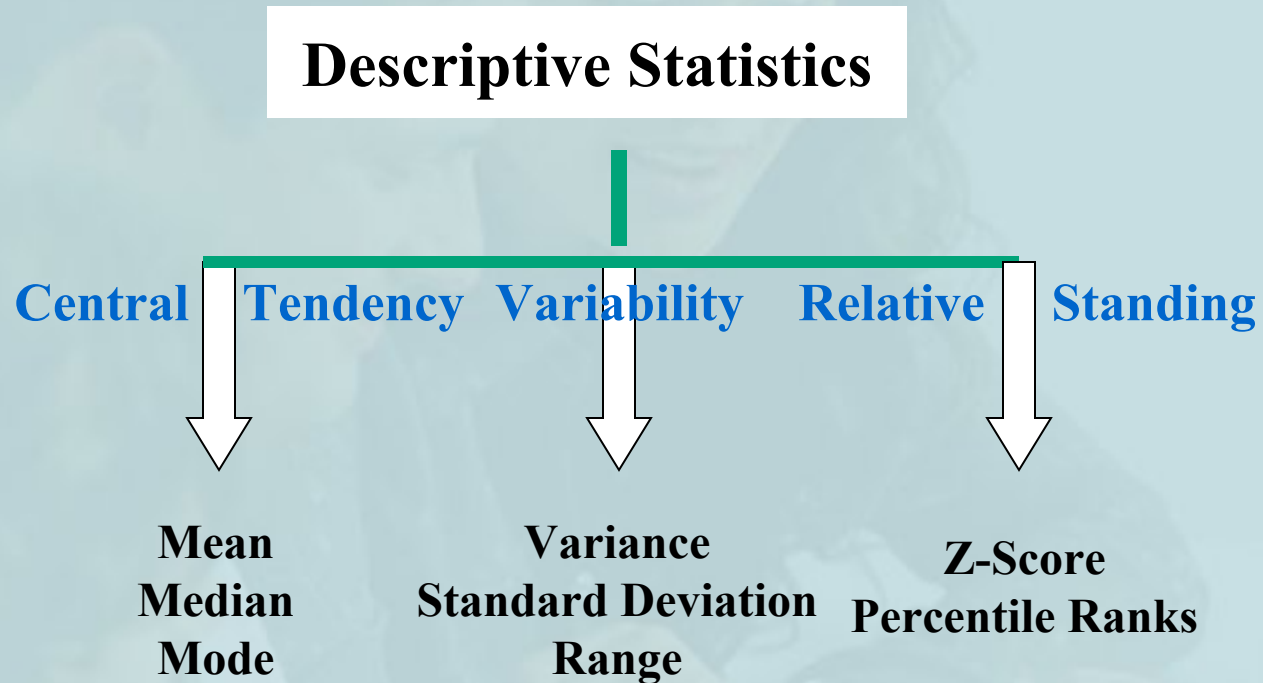
- Statistical Package for Social Sciences (SPSS) most popular
- Other programs
 - Mini-tab
 - Statview
 - SAS (JMP/JMPIN)
 - StatPac
- Use mainframe, PC or Mac platforms

How do you analyze the data?

Conduct descriptive analysis

- Conduct descriptive analysis
 - Measures of central tendency
 - Measures of variability
 - Measures of relative standing

Run Descriptive statistics



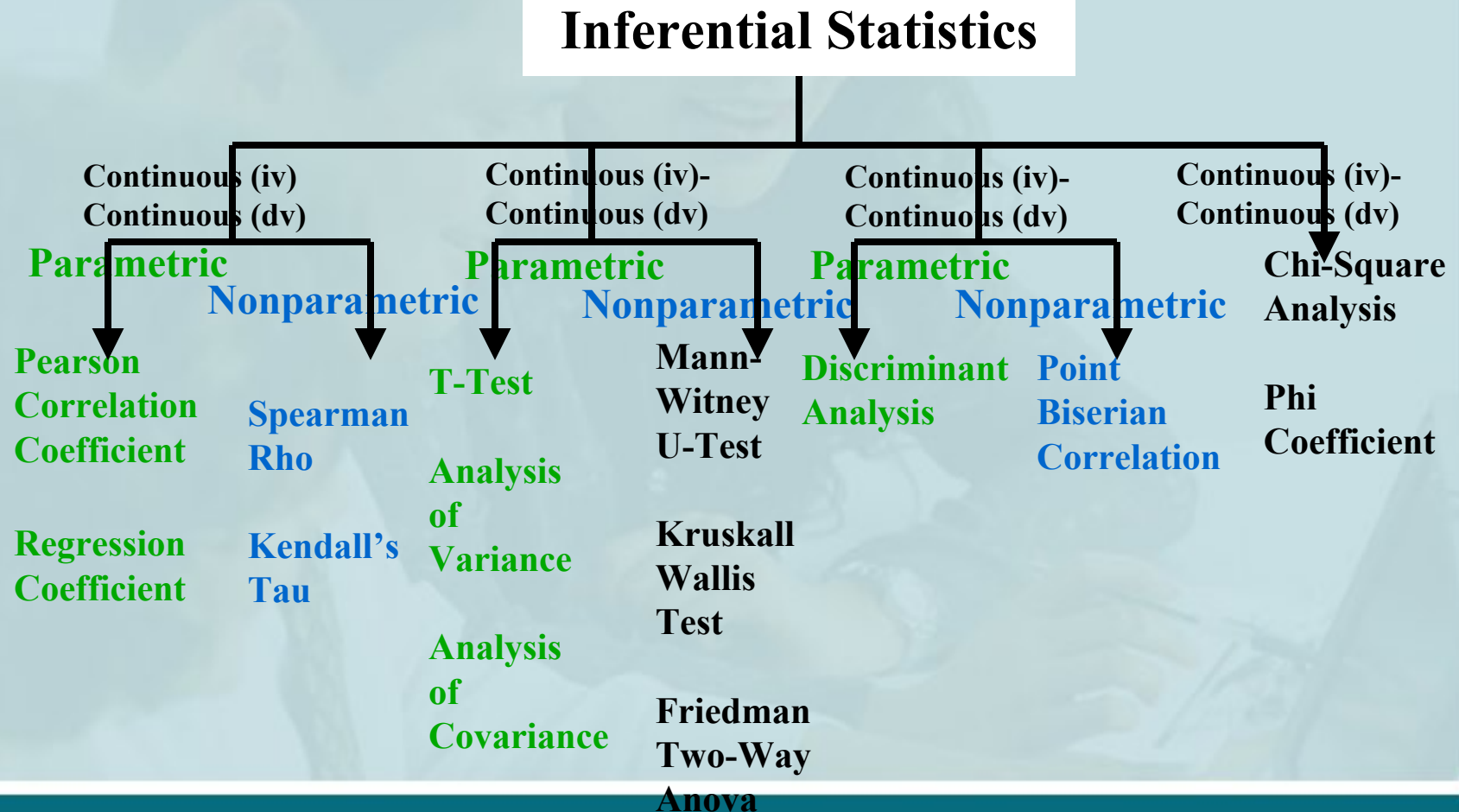
Conduct inferential analysis

- Hypothesis testing
- Confidence interval
- Effect size

Conduct hypothesis tests

- Identify a null and alternative hypothesis
- Set the level of significance (alpha level) for rejecting the null hypothesis
- collect data
- Compute the sample statistic
- Make a decision about rejecting/failing to reject

Computing the sample statistic



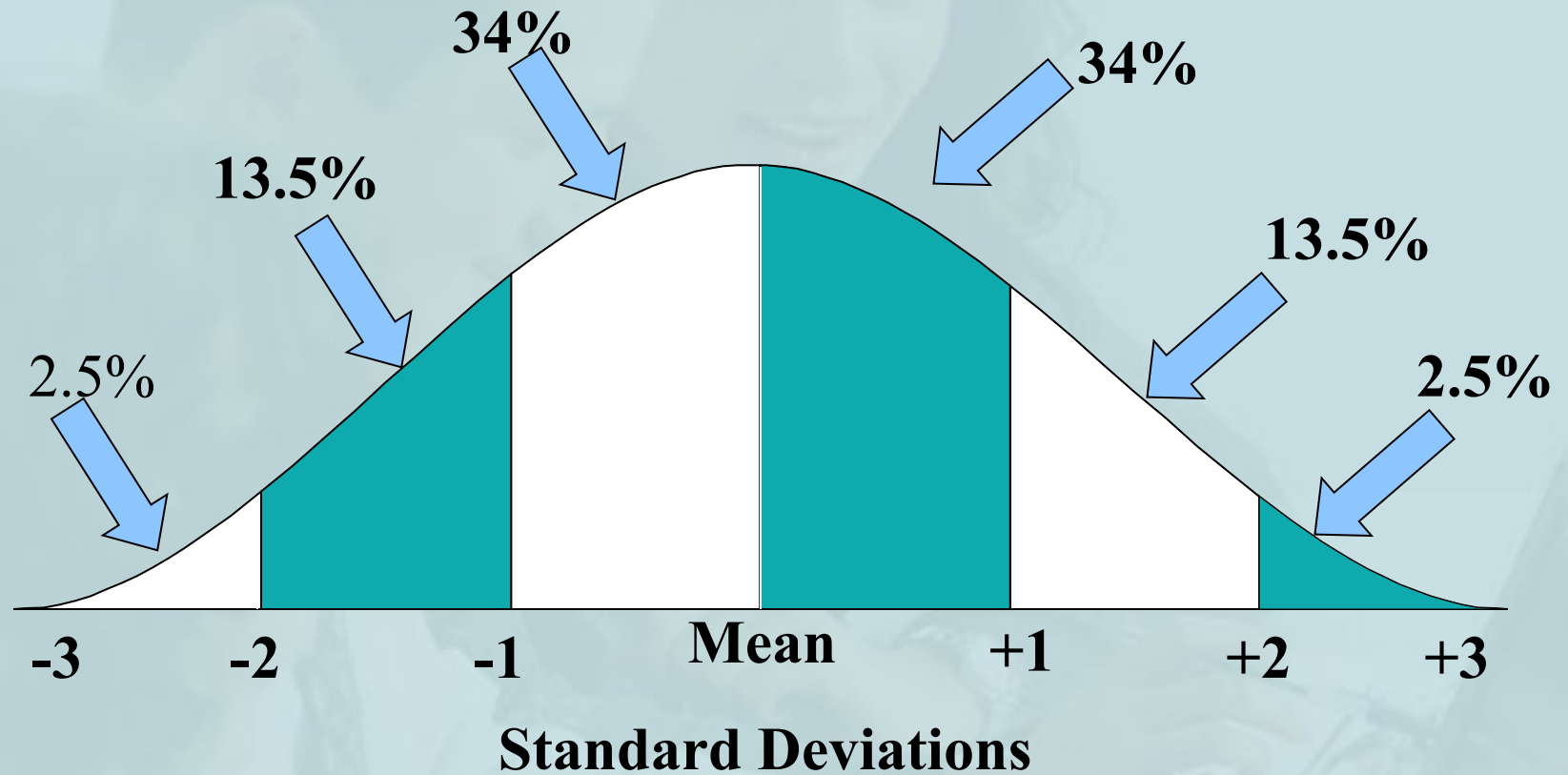
How to select an appropriate statistic (see Table 7.5)

- Determine the type of quantitative research question or hypothesis you want to analyze
- Identify the number of independent variables
- Identify the number of dependent variables
- Identify whether covariates and the number of covariates are used in the research question or hypothesis

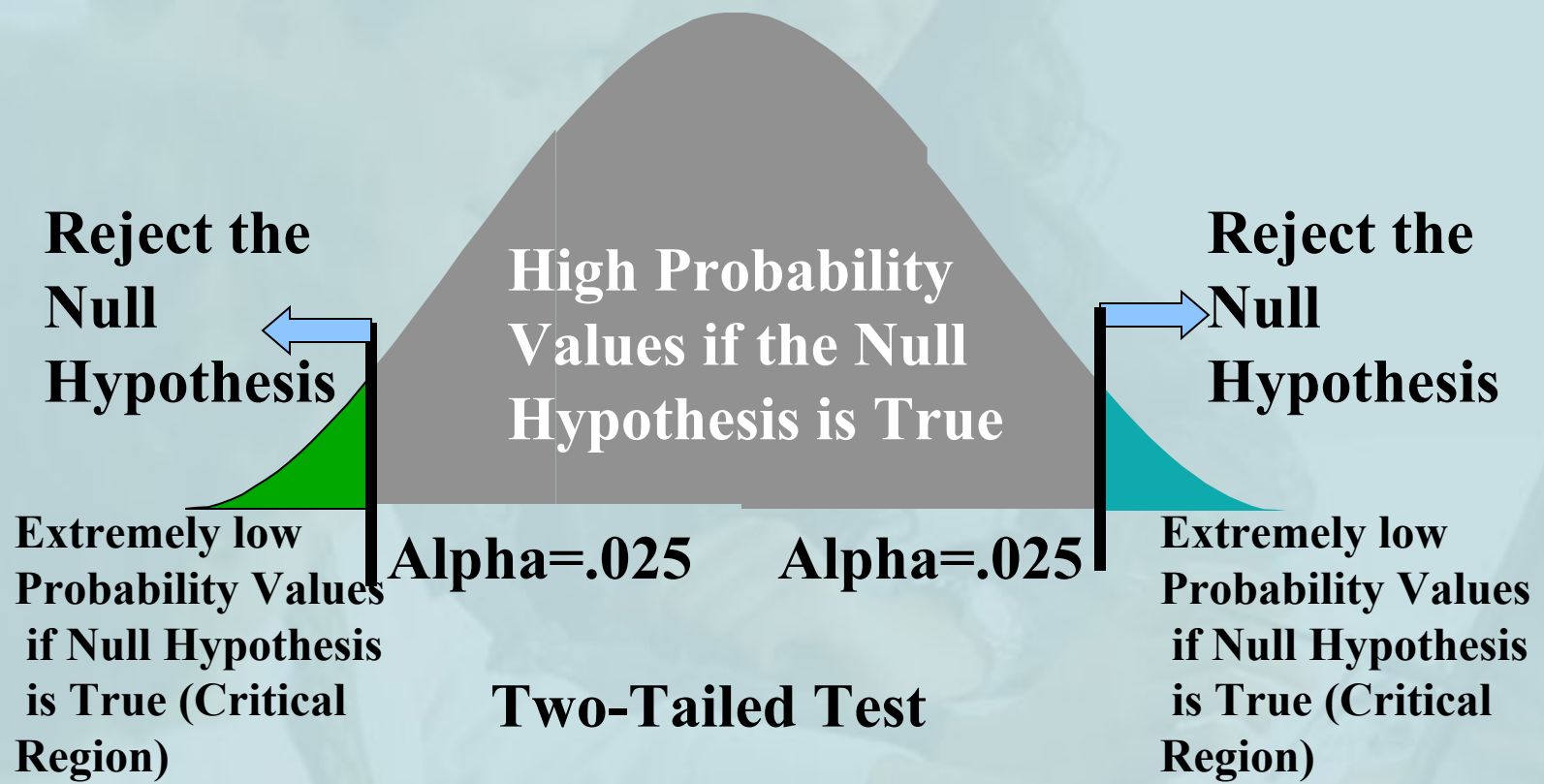
How to select an appropriate statistic (see Table 7.5)

- Consider the scale of measurement for your independent variable(s) in the research question or hypothesis
- Identify the scale of measurement for the dependent variables (e.g. continuous or categorical)
- Determine if the distribution of the scores is normal or skewed

Normal Curve



The Normal Curve of Means Differences of All Possible Outcomes if the Null Hypothesis is True



Outcomes of hypothesis testing: Type I and type II errors

Decision made by
the Researcher
Based on the
Statistical Test
Value

**Reject the
Null
Hypothesis**

**Fail to Reject the
Null Hypothesis**

State of Affairs in the Population

**No Effect-
Null True**

Type I Error
(false positive)
(probability=
Alpha)

Correctly not
rejected
no error

**Effect Exists
Null False**

Correctly
rejected
no error
(probability=
power)

Type II Error
(false negative)
(probability=
Beta)

Other inferential tests

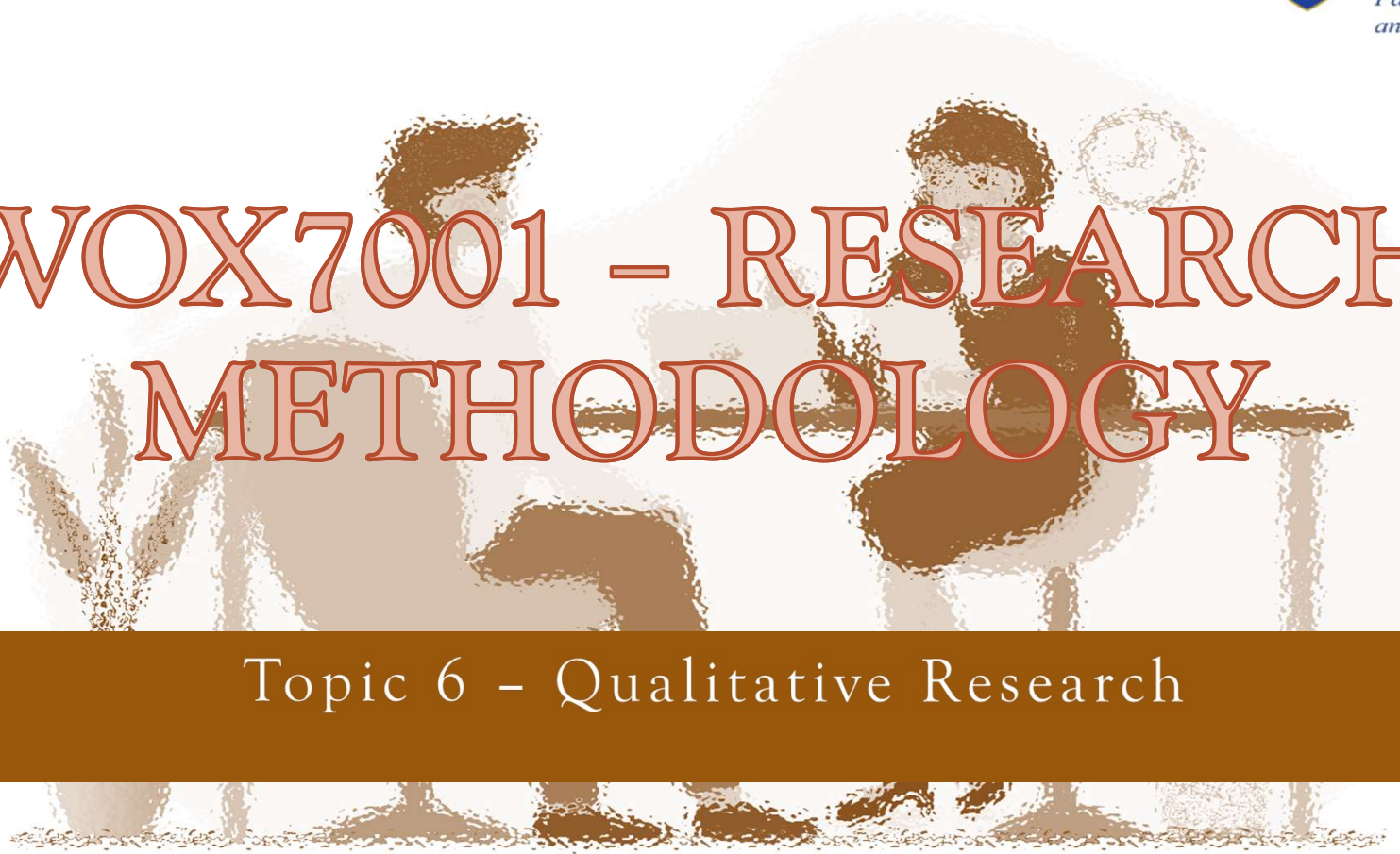
- Confidence intervals
- Effect sizes

How do you report the results?

- Tables summarize statistical information
- Figures (charts, pictures, drawings) portray variables and their relationships
- Detailed explanations about statistical results

How do you discuss the results?

- Summarize major results
- Explain why they occurred
- Advance limitations
- Suggest future research



WOX7001 – RESEARCH METHODOLOGY

Topic 6 – Qualitative Research

Agenda

01

PARTICIPANTS

02

INFORMATION

03

RESEARCH
DATA AND
INSTRUMENT

04

CLASS
ACTIVITIES

Population and Sample

Random “Quantitative” Sampling

Select Representative individuals



To generalize from sample to population



To make claims about the population

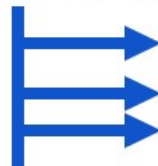
To build/test “theories” that explain the population

Purposeful “Qualitative” Sampling

Select people/sites who can best help us understand our phenomenon



To develop detailed understanding

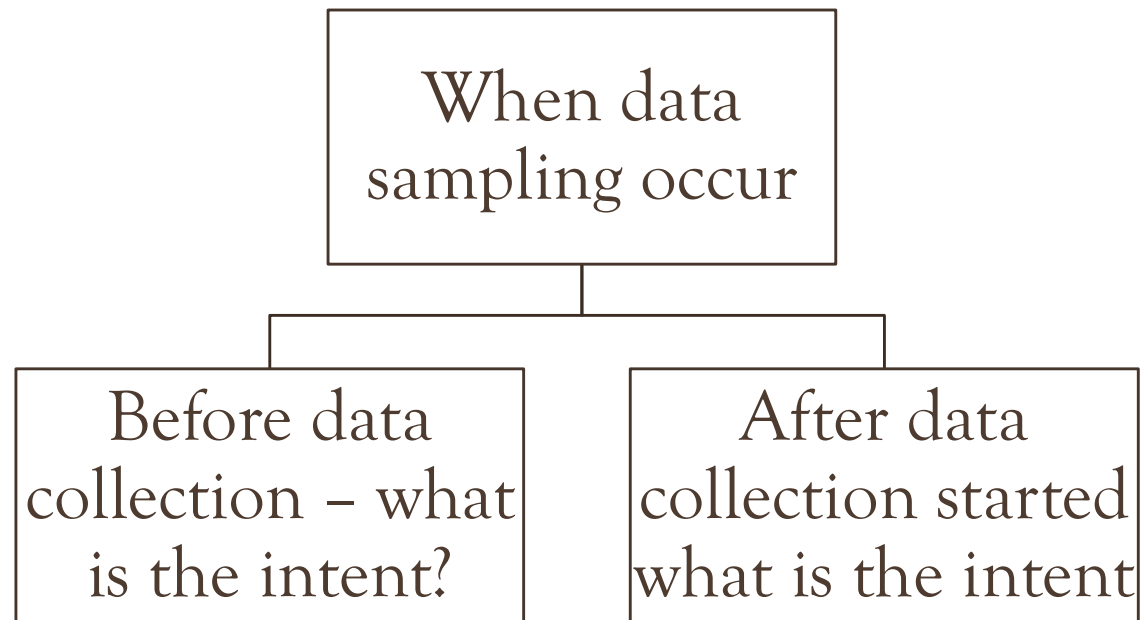


That might be “useful”: information

That might help people “learn” about the phenomenon

That might give voice to “silenced” people

Purposeful Sampling



Before Data Collection

To develop many perspective	Maximal variation sampling	A researcher is conducting a door to door survey to find attitudes towards single parents.
To describe particularly Troublesome or enlightening cases	Extreme case sampling	if you were studying inner city violence, you could study a city with high violence and compare it to a city with low violence
To describe what is “typical” to those unfamiliar with the case	Typical sampling	spending habits of a city in a middle-class suburb (typical) rather than from a poor or rich suburb
To generate a theory or concept	Theory/concept sampling	Grounded theory
To describe some sub-group in depth	Homogeneous sampling	people in a homogeneous sample might share the same age, location or employment.

After Data Collection

To take advantage of whatever case unfolds	Opportunistic sampling	standing on the street asking passers by to join the research.
To locate people or sites to study	Snowball sampling	a researcher who is seeking to study leadership patterns could ask individuals to name others in their community who are influential.
To explore confirming or disconfirming cases	Confirming/disconfirming sampling	Confirmatory cases are additional examples that fit already emergent patterns to add richness, depth and credibility. Disconfirming cases are a source of rival interpretations as well as a means for placing boundaries around confirmed findings

Sample Size

Small for in-depth perspective

1 individual

4 cases

20-30 interviews

What permission are needed

Gain permission from Institutional Review Board (IRB)

Gain permission from “gatekeepers” at the research site

Gatekeepers: individuals at the site who provide site access, help researcher locate people and identify places to study. The gatekeeper may require written permission about the project

Think of some sample of research
gatekeeper

Information (Qualitative)

Prepared by: HSM, NAG and SUH

Information for the gatekeeper

Why their site was chosen

What time and resources are required

What will be accomplished at the site

What potential there is for your presence to be disruptive

What individuals at the site will gain from the study

How you will use and report the results

What Information will you collect?

Observation

Interviews

Documents

Audio-video materials

Observations

An Observation is the process of gathering first-hand information by observing people and places at a research site.

Observational roles

- Participant observer

- Non-Participant observer

Observational roles can be changed

Observations

Conduct multiple observations

Record both descriptive and reflective field notes during the observation

Descriptive field notes describe the events, activities and people

Reflective field notes record personal reflections that relate to their insights, hunches or broad themes that emerge

Interviews

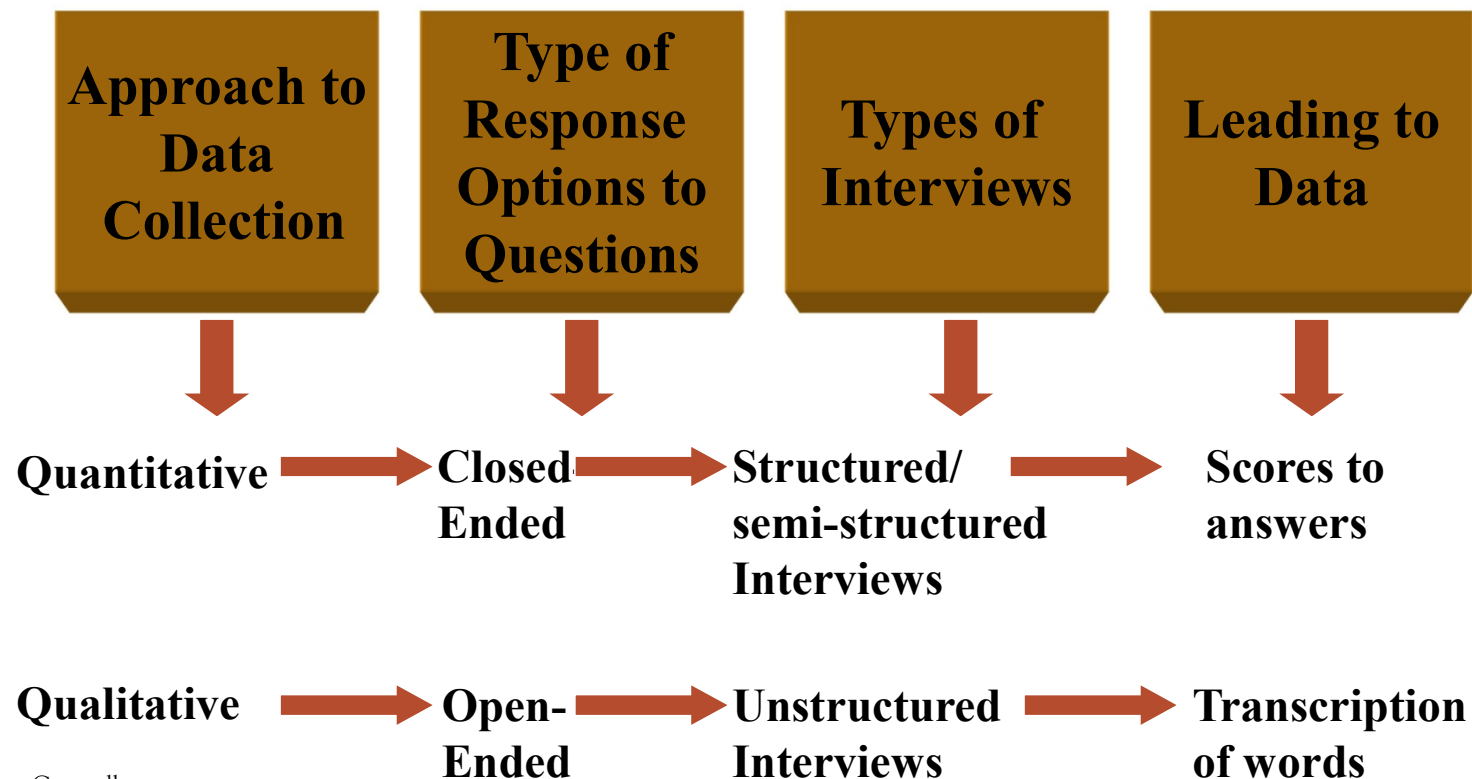
Types: Personal, Phone, e-mail, Focus Group

General open-ended questions are asked

- allows the participant to create options for responding
- participants can voice their experiences and perspectives

Information is recorded then transcribed for analysis

Structured, unstructured, and semi-structured interviews



Documents

Public and private records

Good source for text data

You must obtain permission before using documents

Optically scan documents when possible

Audio-Visual materials

Determine the material that can provide evidence to address your research questions

Determine if the material is available and obtain permission to use it

Check the accuracy and authenticity of the material if you do not record it yourself

Collect the data and organize it

Stopped here

How will you record data?

Using protocols

Observation protocols

Interviewing protocols

Interview protocols

The header: essential information about the interview

Open-ended questions include

- “ice-breaker”

- ones that address major research questions

- probes that clarify and elaborate

Closing comments thanking the participant

Observational protocols

The header: essential information about the interview

Left column to record descriptive notes

Right column to record reflective notes

A picture of the site may be sketched

How do you administer data collection?

Time needed for data collection

- Limit initial collection to one or two observations or interviews

- Time is needed to establish a substantial data base

Obtaining permission to use materials

Ethical issues

- Anonymity of participants

- Convey true purpose of study without deception

Research Data and Instrument

Prepared by: HSM, NAG and SUH

Research Instrument

An instrument is a tool for measuring, observing, or documenting qualitative

Types of Instruments

- Interview questionnaire

- Checklist for observation

- Document analysis

A SNIPPET OF RESPONSES TO ONE INTERVIEW QUESTION POSED TO THREE RESEARCH PARTICIPANTS

Researcher question (open ended): I understand all of you have been teaching high school continuously for 40 years or more. What advice do you have for teachers beginning their first year of teaching?

Teacher 1: I was scared the first year or two, almost every day. Not for my safety or anything like that, but I kept thinking I might fail, maybe my students wouldn't learn anything. I was not that much older than my students. Now, of course, I'm more like a grandmother figure to them. I was worried they would not take me seriously—you know, cut up and take advantage of a young, first-year teacher. Maybe my insecurity showed, maybe they saw my lack of confidence or figured I didn't know what I was doing because I was not very experienced. Now, of course, I think how silly to have worried about such things. So, to answer your question, I would say, "Don't worry, be happy."

Code: Overcome insecurity

I know some people say "Forget everything you learned in your teacher prep program [pause], real learning goes on in the trenches." Sure, you learn by doing, but some old ideas in teaching textbooks are time honored and priceless.

Code: Use learning theory

I recall one theory that focused on time, you know, all kids can learn if given enough time. That's so true. If new teachers know that fact, then they won't get frustrated. They will know that learning algebra, for example, will eventually happen with more time, maybe after school, maybe during the summer. New teachers have to know some kids learn really fast; others, really slow. But they do learn. But there is a clock with a buzzer, so I know time runs out. Before time runs out, the teacher should try something new, maybe explaining it with a sketch on paper—yeah, that's it. Try something new.

Code: Experiment with methods

Teacher 2: I remember I marched into my classroom full of vigor and ready to conquer the world. Boy, did those rascals put me in my place! I remember that I thought about quitting my whole first year, crying on the way home sometimes. My dad was a teacher, and he kept saying the first year is hard—just keep going, he would say. That was hard. [Now, what was your question? Laugh.] Oh yeah, I would tell new teachers that it gets better and better every year, like a fine wine! If it didn't, then why would I stay in the classroom for 40 years!

Code: Brace yourself; it only improves

They have to know that the first year is trial and error. Well, not just the first year; you have to experiment all the time to find the right approach with some students.

Code: Experiment with methods

They should know that you won't be teaching the same way year after year. You can't be a repeating robot. People change with the times; every year can be different. What is that motto? Change is good, or something like that. Sometimes you have to be creative.

Code: Be flexible; adapt to change

I used to complain about chalk on my clothes; now I complain about lugging my laptop to school. You never know when the school's computer—I think it's a 1988 Apple II or something—will crash on you. I use my computer a lot to update grades, make changes to assignments and all that. My students can go to a website 24/7 and learn what to do for homework. So, I guess my advice is to roll with the punches and don't expect a routine. Routines get boring after a while anyway. Yeah, I would say keep changing with the times and welcome new ways of doing things.

Code: Welcome innovation

Everything changes so fast these days anyway.

Teacher 3: I would say prepare to wear many hats. That is because teaching today is very different than when I first started. I would say something like, you are a part-time teacher, part-time counselor, part-time social worker, part-time therapist, even part-time parent! Teaching in the old days was pretty much teaching; today it is far more and new teachers must be prepared for that.

Code: Brace yourself for multiple roles

I don't think they train new teachers to be social workers, but what else can you do when a student comes in hungry and holds back tears? What did she just experience? What do you do when another student comes in acting high on drugs? You see, teaching becomes harder and harder when you know that some students cannot concentrate on learning.

Code: Be prepared for challenges beyond teaching

Many have huge problems that they deal with. I do what I can, but with so many other students, it's just a hard job. I think they call it compassion fatigue, or something like that. I'm one of the lucky ones; I can go to the mountain cabin for most of the summer. Others, I know, take on other jobs during the summer to pay bills. New teachers should know about the challenges from Day 1, challenges that are not related to lesson plans or technology. The problems are not insurmountable. If they were, I would have started a business or something like that instead. I've loved every class, every semester, because you can make a difference in kids' lives. Students comes back sometimes.

Instrument Open ended questions - sample

Data interpretation strategies

Method 1: Hierarchy

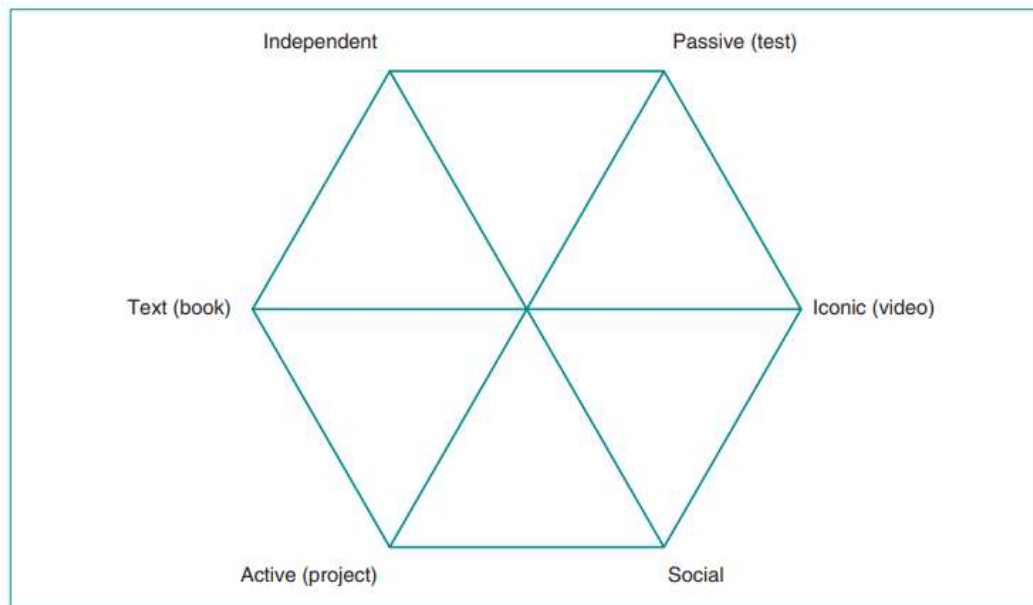
Example after interview the answers transforms into hierarchy as follows. The samples shows Parents' Attitudes Toward Education Displayed as a Hierarchy

Higher Priority			Lower Priority		
Economic	Tradition	Self-Efficacy	Economic	Tradition	Self-Efficacy
Escape Thrive	Continuity Values	Ability Personality	Futility Survive	Expectation Information	Barriers Helplessness

Data interpretation strategies

Method 2: Taxonomies

Example after interview the answers transforms into Taxonomies as follows. The samples shows A hypothetical example of different styles of online learning



Data interpretation strategies

Method 3: Network

Example after interview the answers transforms into networks as follows. Network represent the organizational systems that reveal connections with a process that occurs over time. The sample shows the A hypothetical example of a network that reveals linkages in the process of dropout.



Data interpretation strategies

Method 4: Tables and Cross Tabulations

Example after interview the answers transforms into cross tabulations as follows. The sample shows the Hypothetical Cross Tabulation of Type of Dishonesty and Type of Course.

Type of Dishonesty	Type of Course	
	F2F	Online
Social	Lower	Higher
Nonsocial	Same	Same

THANK YOU

Prepared by: HSM, NAG and SUH

Chapter 8

Collecting Qualitative Data



Key Ideas

- What participants and sites will be studied?
- What information will you collect?
- How will you record data?
- How will you administer data collection?

Who will be studied?

Purposeful Sampling

Random “Quantitative” Sampling

Select Representative individuals

- To generalize from sample to population
 - To make claims about the population
 - To build/test “theories” that explain the population

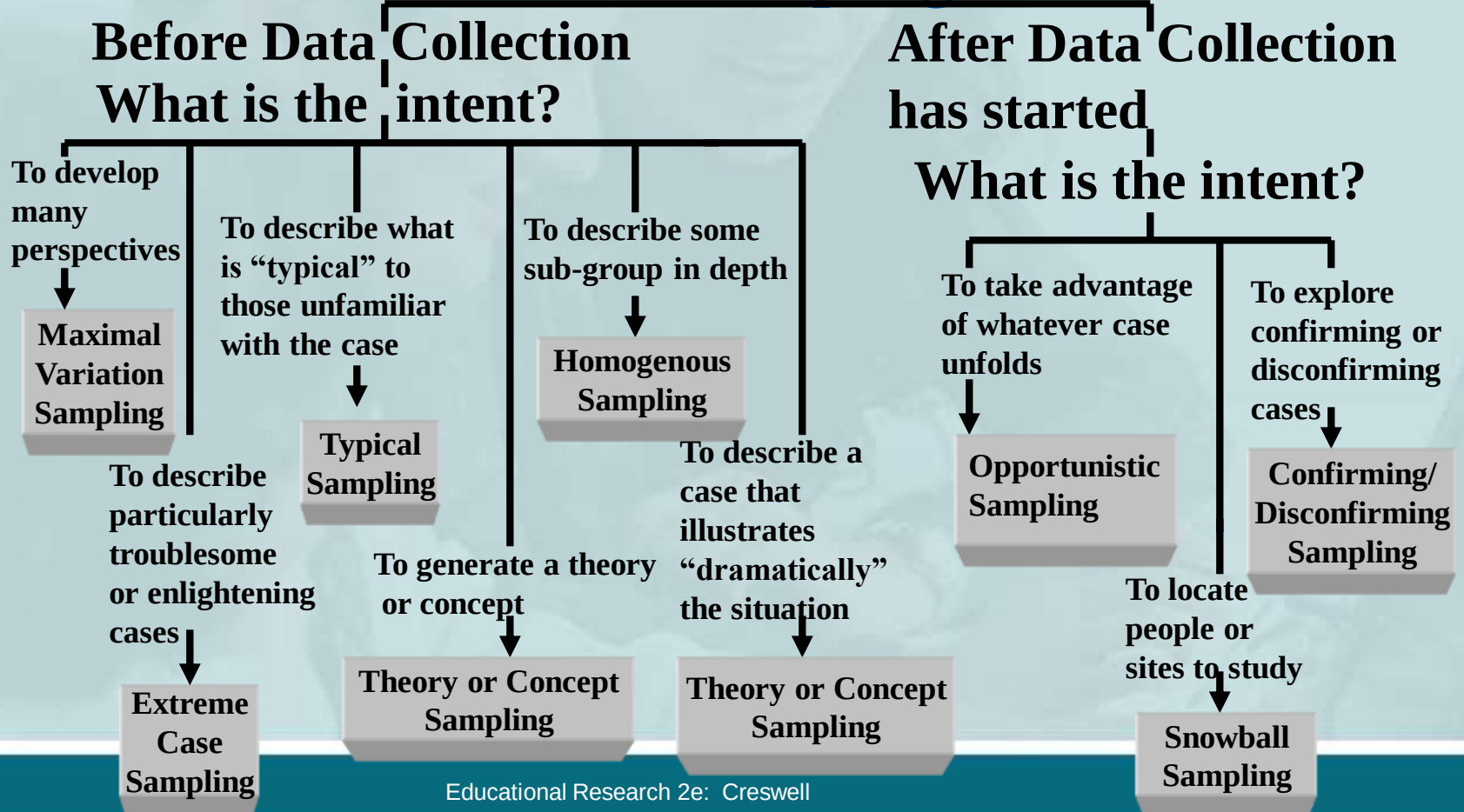
Purposeful “Qualitative” Sampling

Select people/sites who can best help us understand our phenomenon

- To develop detailed understanding
 - That might be “useful”: information
 - That might help people “learn” about the phenomenon
 - That might give voice to “silenced” people

Types of purposeful sampling

When Does Sampling Occur?



Sample size

- Small for in-depth perspective
 - 1 individual
 - 4 cases
 - 20-30 interviews

What permissions are needed?

- Gain permission from Institutional Review Board (IRB)
- Gain permission from “gatekeepers” at the research site
 - Gatekeepers: individuals at the site who provide site access, help researcher locate people and identify places to study
 - The gatekeeper may require written permission about the project

Differences between qualitative and quantitative permissions

- Study conducted at site
- In-depth interviewing and prolonged observing
- Personal nature of some qualitative studies (not amenable to variables and measures)

Information for the gatekeeper

- Why their site was chosen
- What time and resources are required
- What will be accomplished at the site
- What potential there is for your presence to be disruptive
- What individuals at the site will gain from the study
- How you will use and report the results

What information will you collect?

- Observations
- Interviews
- Documents
- Audio-Visual Materials

Observations

- An Observation is the process of gathering first-hand information by observing people and places at a research site.
- Observational roles
 - Participant observer
 - Non-Participant observer
- Observational roles can be changed

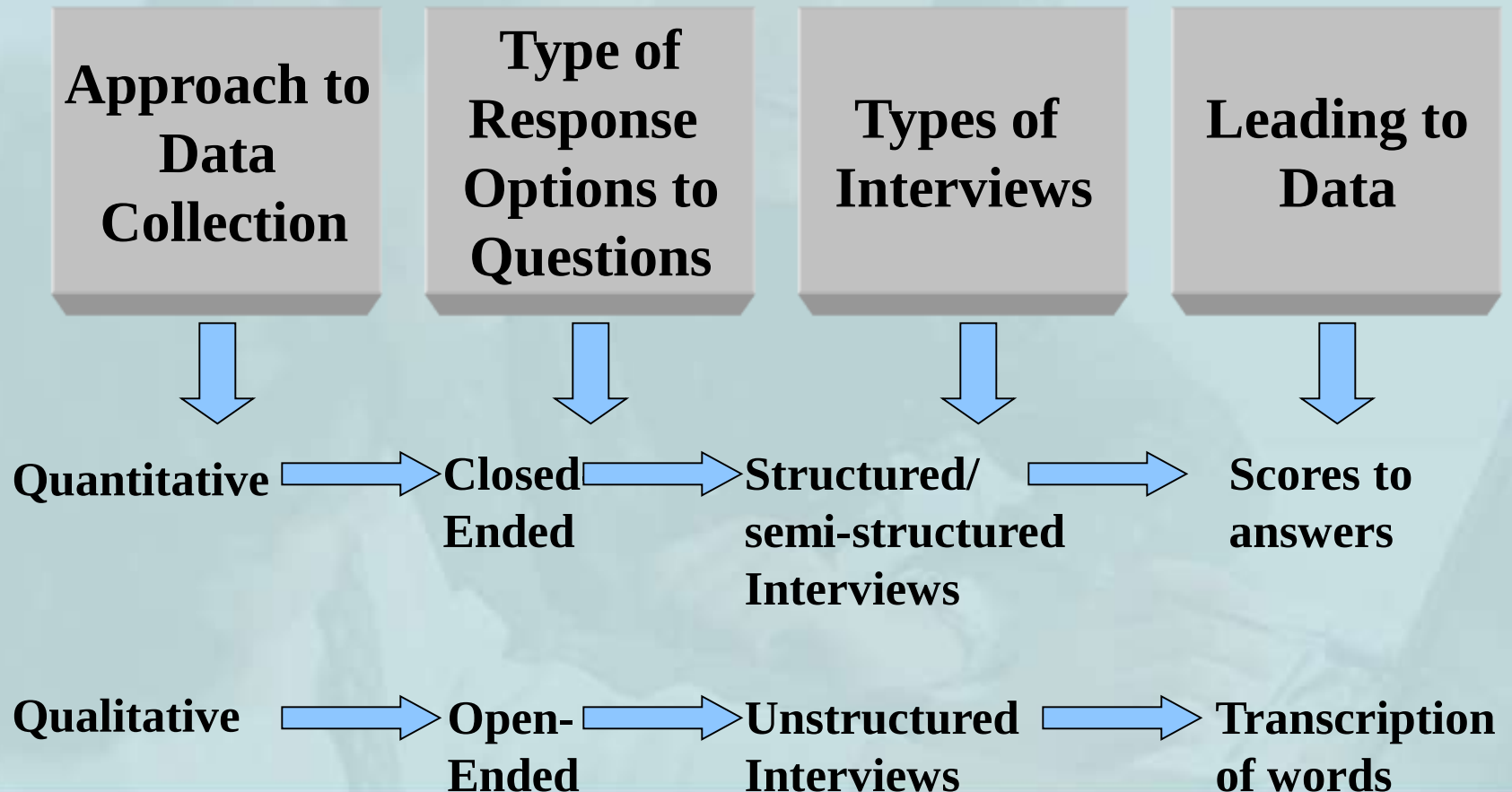
Observations

- Conduct multiple observations
- Record both descriptive and reflective field notes during the observation
 - Descriptive field notes describe the events, activities and people
 - Reflective field notes record personal reflections that relate to their insights, hunches or broad themes that emerge

Interviews

- Types: Personal, Phone, e-mail, Focus Group
- General open-ended questions are asked
 - allows the participant to create options for responding
 - participants can voice their experiences and perspectives
- Information is recorded then transcribed for analysis

Structured, unstructured, and semi-structured interviews



Documents

- Public and private records
- Good source for text data
- You must obtain permission before using documents
- Optically scan documents when possible

Audio-Visual materials

- Determine the material that can provide evidence to address your research questions
- Determine if the material is available and obtain permission to use it
- Check the accuracy and authenticity of the material if you do not record it yourself
- Collect the data and organize it

How will you record data?

- Using protocols
 - Observation protocols
 - Interviewing protocols

Interview protocols

- The header: essential information about the interview
- Open-ended questions include
 - “ice-breaker”
 - ones that address major research questions
 - probes that clarify and elaborate
- Closing comments thanking the participant

Observational protocols

- The header: essential information about the interview
- Left column to record descriptive notes
- Right column to record reflective notes
- A picture of the site may be sketched

How do you administer data collection?

- Time needed for data collection
 - Limit initial collection to one or two observations or interviews
 - Time is needed to establish a substantial data base
- Obtaining permission to use materials
- Ethical issues
 - Anonymity of participants
 - Convey true purpose of study without deception

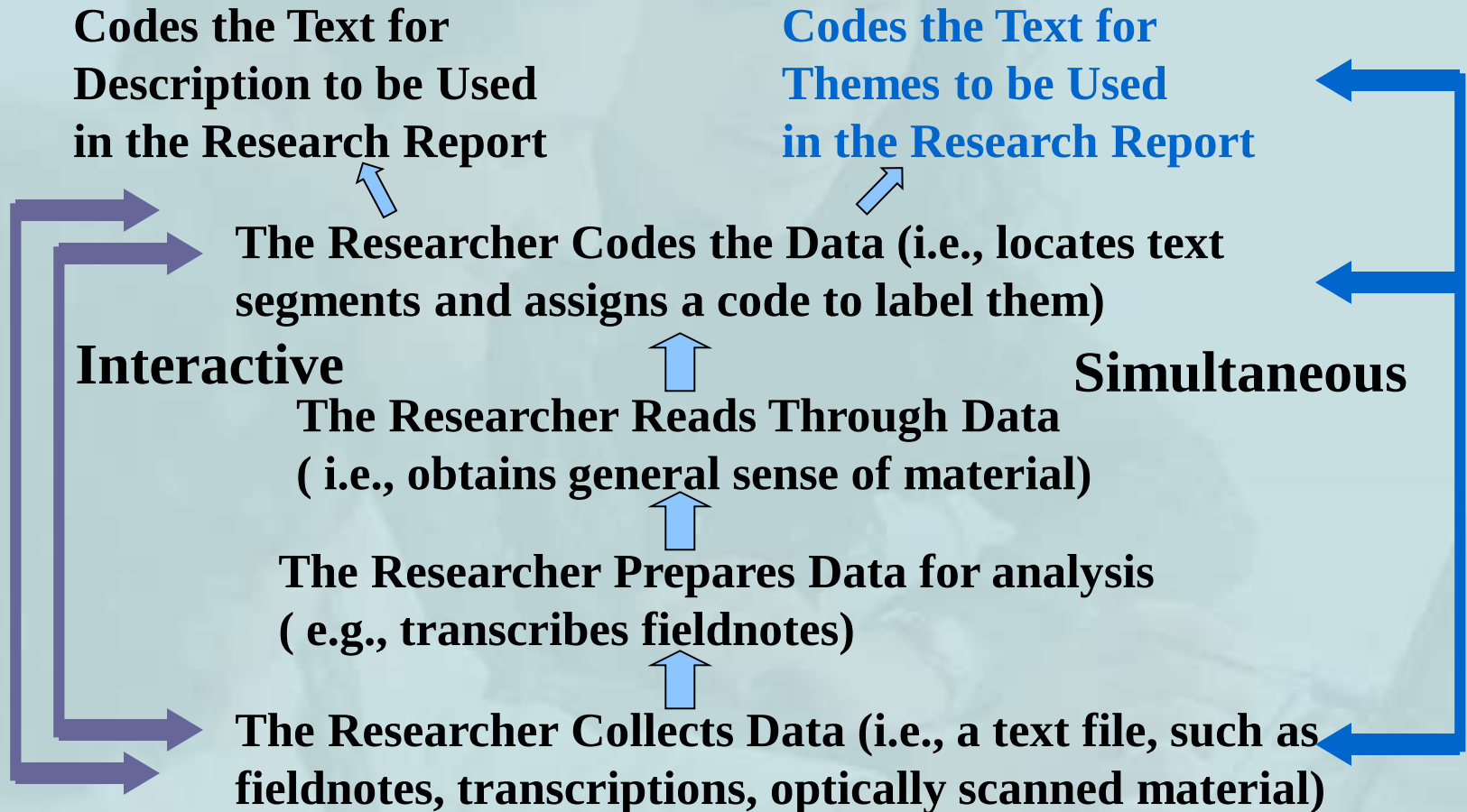
Analyzing and Interpreting Qualitative Data



Key Ideas

- Prepare and organize the data for analysis
- Explore and code the data
- Develop description and themes from the data
- Represent and report the findings
- Interpret the findings
- Validate the accuracy and credibility of the findings

What is the Process of Data Analysis?



How do You Prepare and Organize the Data?

- Develop a matrix or table of sources that can be used to organize the material
- Organize material by type
- Keep duplicate copies of materials
- Transcribe data
- Prepare data for hand or computer analysis (and select computer program)

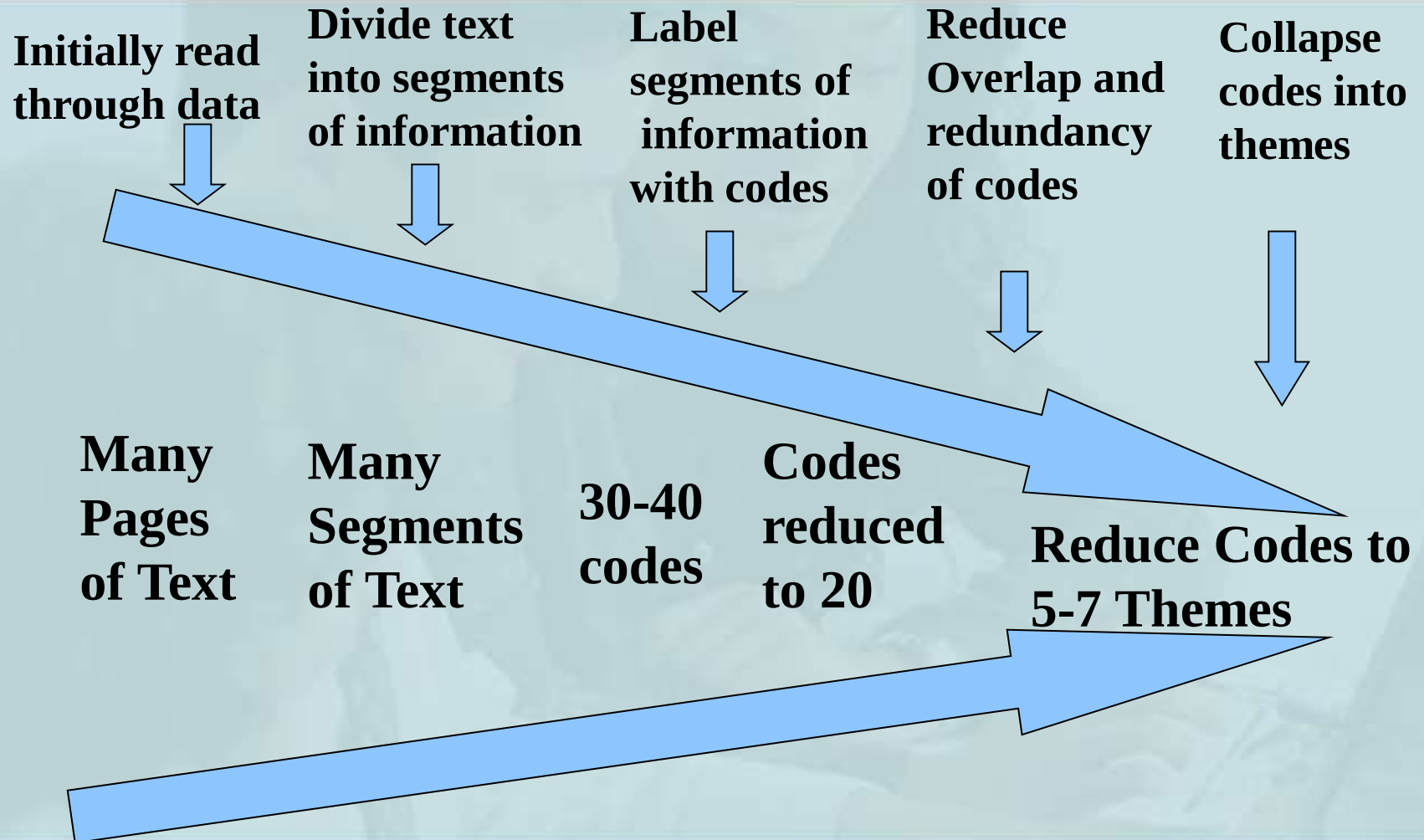
How do you Explore and Code the Data?

- Obtain a general sense of the data by performing a preliminary exploratory analysis
- Memo ideas
- Consider whether more data are needed
- Coding the data

Steps in Coding

- Read through all transcripts
- Start with one transcript
- Identify text segments – ask “what is this person saying?”
- Bracket text segment
- Assign code word
- Reduce redundancy
- Collapse codes into themes

A Visual Model of the Coding Process in Qualitative Research



How do You Use Codes to Build Description?

- Describe
 - People
 - Events
 - Activities
 - Processes
- Describe in detail

Coding Used in a Descriptive Passage

The Incident and Response

Description builds
from broad to narrow →

Situate the reader
in the place →

Provide details →

Detail to create a
sense of “being there” →

Use of action verbs
and vivid modifiers
and adjectives →

The incident occurred on the campus of a large public university in a Midwestern city. A decade ago, this city had been designated an “all-American city,” but more recently, its normally tranquil environment has been disturbed by an increasing number of assaults and homicides. Some of these violent incidents have involved students at the university.

The incident that provoked this study occurred on a Monday in October. A forty-three-year-old graduate student, enrolled in a senior-level actuarial science class, arrived a few minutes before class, armed with a vintage Korean War military semiautomatic rifle loaded with a thirty-round clip of thirty caliber ammunition. He carried another thirty-round clip in his pocket. Twenty of the thirty-four students in the class had already gathered for class, and most of them were quietly reading the student newspaper. The instructor was en route to class.

The gunman pointed the rifle at the students, swept it across the room, and pulled the trigger. The gun jammed. Trying to unlock the rifle, he hit the butt of it on the instructor’s desk and quickly tried firing it again. Again it did not fire. By this time, most students realized what was happening and dropped to the floor, overturned their desks, and tried to hide behind them. After about twenty seconds, one of the students shoved a desk into the gunman, and students ran past him out into the hall and out of the building. The gunman hastily departed the room and went out of the building to his parked car, which he had left

How do You Use Codes to Identify Themes?

- Ordinary themes
- Unexpected themes
- Social science themes
- Layering and connecting themes

Coding Used in Theme Passage

Title for theme
based on words of
participant

Safety

Evidence for themes
based on multiple
perspectives of
participants

Within themes
are sub-themes

The violence in the city that involved university students and the subsequent gun incident that occurred in a campus classroom shocked the typically tranquil campus. A counselor aptly summed up the feelings of many: "When the students walked out of that classroom, their world had become very chaotic; it had become very random, something had happened that robbed them of their sense of safety." Concern for safety became a central reaction for many informants. When the chief student affairs officer described the administration's reaction to the incident, he listed the safety of students in the classroom as his primary goal, followed by the needs of the news media for details about the case, helping all students with psychological stress, and providing public information on safety. As he talked about the safety issue and the presence of guns on campus, he mentioned that a policy was under consideration for the storage of guns used by students for hunting. Within 4 hours after the incident, a press conference was called during which the press was briefed not only on the details of the incident, but also on the need to ensure the safety of the campus. Soon thereafter the university administration initiated an informational campaign on campus safety. A letter, describing the incident, was sent to the university board members. (One board member asked, "How could such an incident happen at this university?")

How do You Represent Findings?

- Comparison table
- Descriptive table
- Hierarchical tree
- Figures/diagrams
- Drawings

How do You Report Findings?

- Multiple perspectives for each theme
- Metaphors and analogies
- Quotes
- Detail
- Tensions and contradictions

How do You Interpret Findings?

- Interpretation is not neutral
- Reflect about the personal meaning of the data
- Compare and contrast personal viewpoints with the literature
- Address limitations of the study
- Make suggestions for future research

How do You Validate the Accuracy of Your Findings?

- Member checking: Members check the accuracy of the account
- Triangulation: Using corroborating evidence
- External: Hiring the services of an individual outside the study to review the study



**UNIVERSITI
MALAYA**

*Faculty of Computer Science
and Information Technology*

WOX7001 - RESEARCH METHODOLOGY

Topic 7 - Mixed Method Research

Agenda

01

DEFINITION

02

DESIGNING
MIXED METHOD
STUDY

03

MIXED METHOD
RQ

How would you combine two types of data?

Qualitative Text Data

- ◆ This is a sample of a text file of words that might be collected on interview transcripts, observation fieldnotes, or optically-scanned documents.

Quantitative Numeric Data

◆ 2 3 4 2 5 2 3 1 1 2 3 4 2 3 3 2
1 1 1 5 3 4 1 2 3 1 4 4 5 5 4 1
2 1 4 3 3 5 1 4 2 3 1 5 5 2 2 1
5 3 5 1 3 1 5 3 2 2 5 1 3 2 4 4
3 1 2 4 2 2 4 1 5 5 4 2 1 5

A mixed methods researcher...

- Collects both quantitative and qualitative data
- “Mixes” them
- “Mixes” them at the same time (concurrently) or one after the other (sequentially)
- Emphasizes both equally or unequally

A definition...

- *Mixed methods research is both a method and methodology for conducting research that involves collecting, analyzing, and integrating quantitative and qualitative research in a single study or a longitudinal program of inquiry.*
- *The purpose of this form of research is that both qualitative and quantitative research, in combination, provide a better understanding of a research problem or issue than either research approach alone.*

Mixing or linking the data

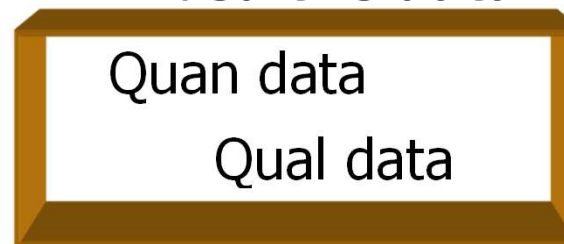
Converge data:



Connect data:



Embed the data:



What is the reason for using mixed methods?

- ◆ The insufficient argument – either quantitative or qualitative may be insufficient by itself
- ◆ Multiple angles argument – quantitative and qualitative approaches provide different “pictures”
- ◆ The more-evidence-the-better argument – combined quantitative and qualitative provides more evidence
- ◆ Community of practice argument – mixed methods may be the preferred approach within a scholarly community
- ◆ Eager-to-learn argument – it is the latest methodology
- ◆ “Its intuitive” argument – it mirrors “real life”

Designing a Mixed Methods Study

- ◆ Preliminary considerations
- ◆ Creating a title
- ◆ Posing a general question
- ◆ Listing the types of data collection and analysis
- ◆ Making explicit your worldview
- ◆ Identifying your research design
- ◆ Drawing a figure of your design
- ◆ Writing a purpose statement
- ◆ Writing research questions
- ◆ Completing a research plan

List your types of data collection (a review)

- ◆ Quantitative data (closed-ended)
 - ◆ Instruments
 - ◆ Behavioral checklists
 - ◆ Records
- ◆ Qualitative data (open-ended)
 - ◆ Interviews
 - ◆ Observations
 - ◆ Documents
 - ◆ Audio-visual materials

Activity – List your sources of data

Quantitative Sources of Data

- ◆ Via questionnaire analysis
- ◆ Via experiments

Qualitative Sources of Data

- Via observation analysis
- Via analysis of interview
transcribe

List your approach to data analysis (a review)

- Quantitative analysis
 - Use statistical analysis,
 - For description
 - For comparing groups
 - For relating variables
 - Design-type
- Qualitative analysis
 - Use text and images,
 - For coding
 - For theme development
 - For relating themes
 - Design-type

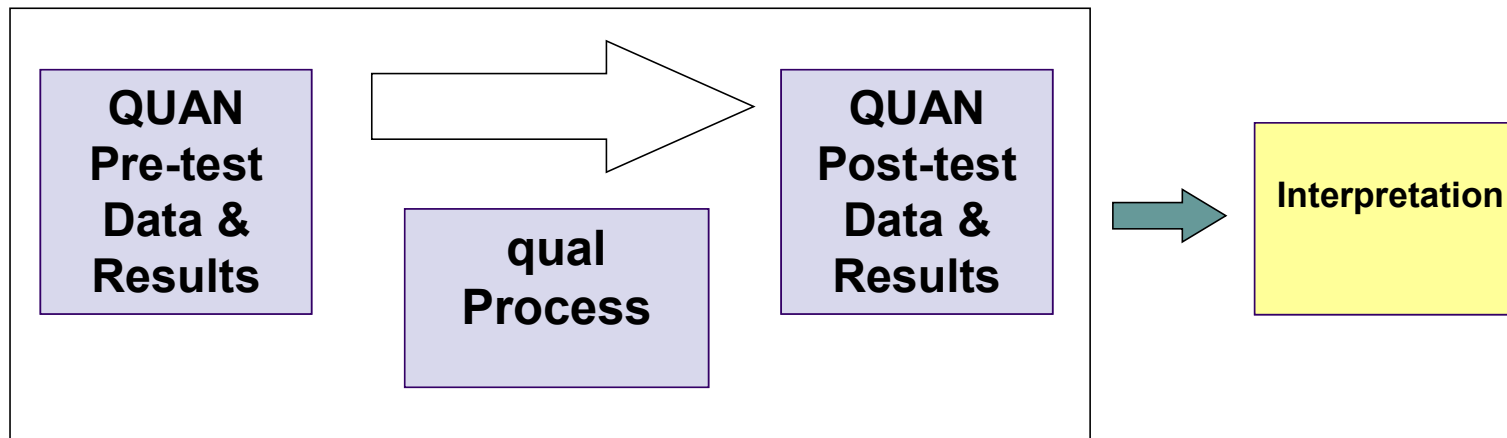
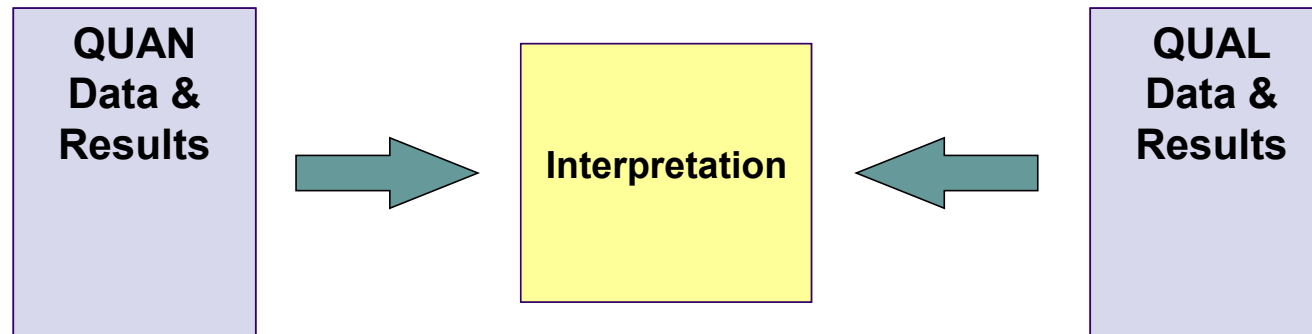
Make explicit your (theory)

- ◇ Components:
 - ◇ What is it
 - ◇ Who has used it in your field
 - ◇ How it will shape your study (rephrase your guiding research question, if a theory applies)

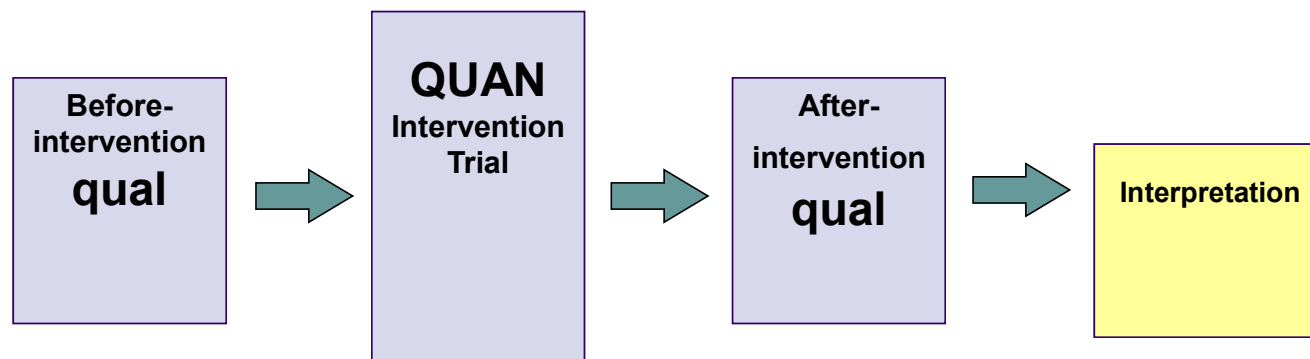
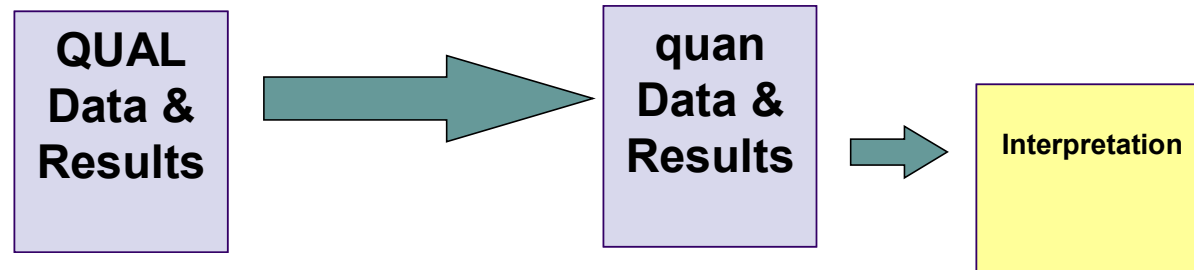
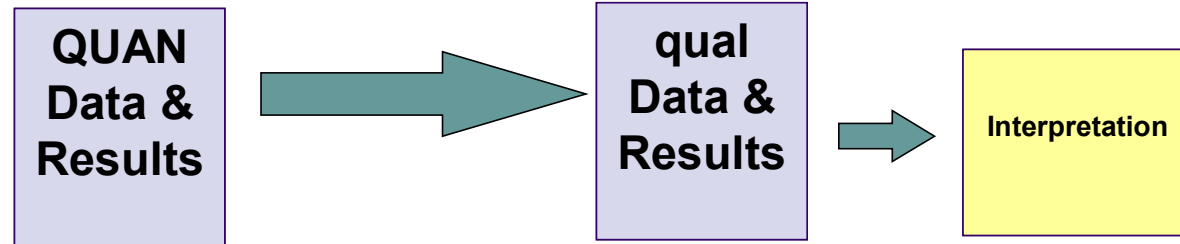
State your procedures (methods)

- ◆ Procedures for handling your qualitative and quantitative data
- ◆ Sequence – concurrent or sequential or both
- ◆ Emphasis – emphasis on qualitative or quantitative
- ◆ Sometimes both concurrent and sequential phases are used
- ◆ Designs may include more than two phases
- ◆ Think about using a simple, elegant design

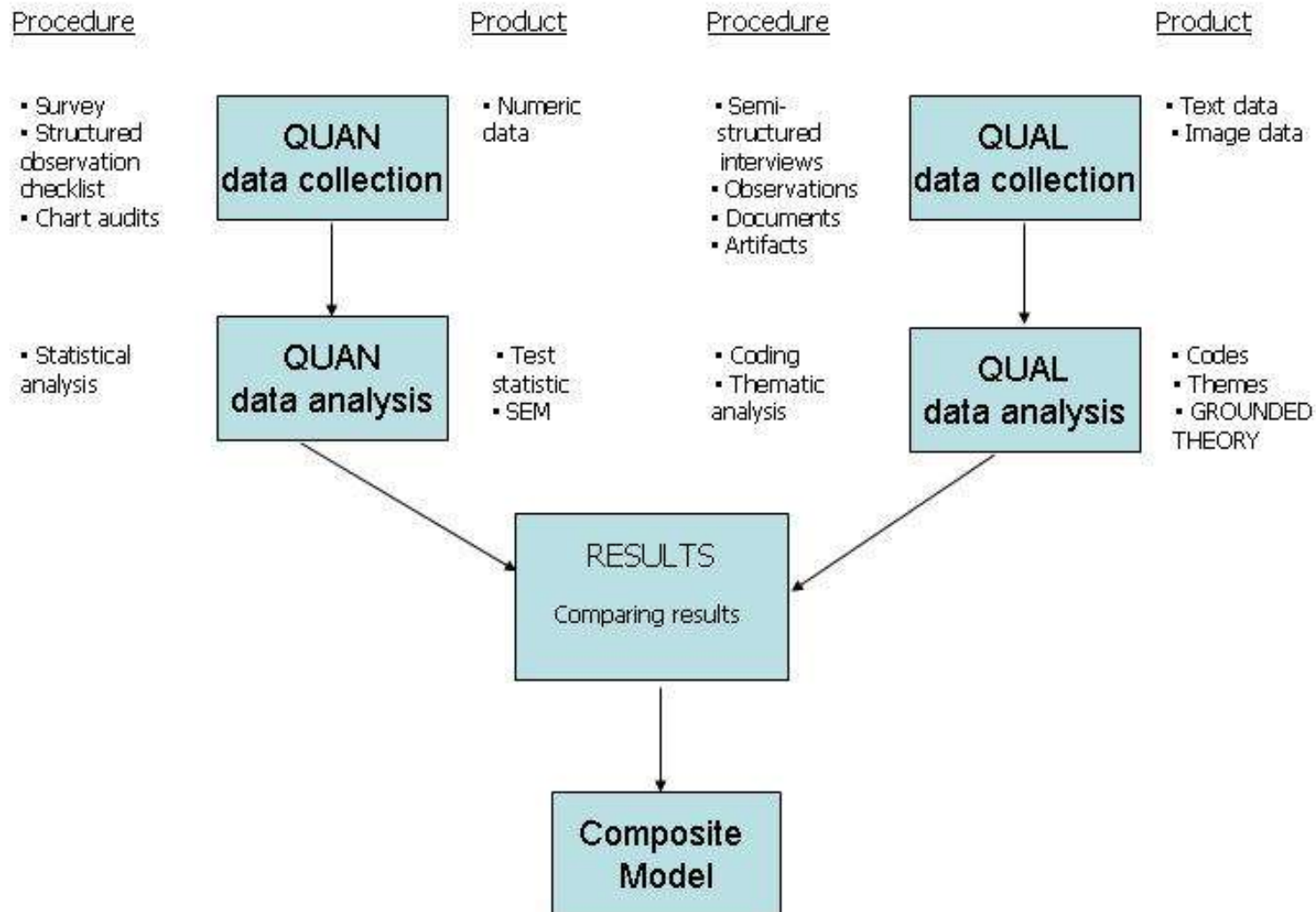
Concurrent Mixed Methods Designs



Sequential designs



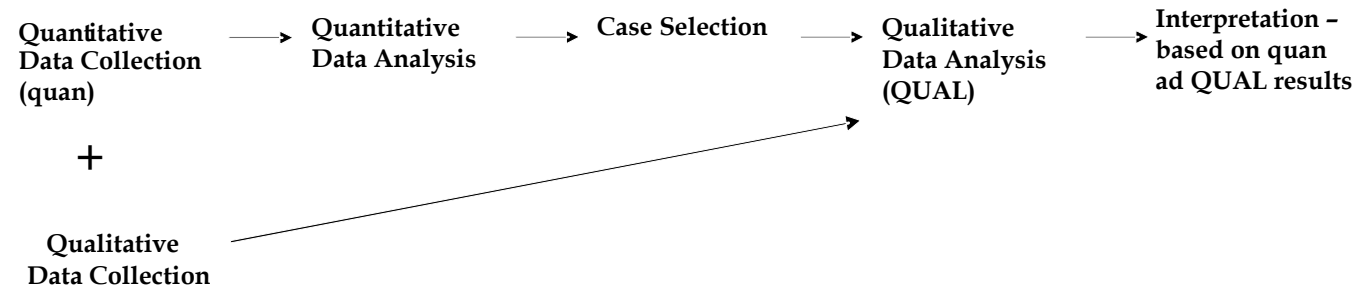
Concurrent Triangulation Design Visual Model



Design options

Design Name	Equal priority	QUAN emphasis	QUAL emphasis
Concurrent, triangulation	QUAL+QUAN	QUAN+qual	QUAL+quan
Concurrent, embedded	n/a	QUAN(qual)	QUAL(quan)
Explanatory, sequential, quan first	QUAN→QUAL	QUAN→qual	quan→QUAL
Exploratory sequential, qual first	QUAL→QUAN	qual→QUAN	QUAL→quan
Sequential, embedded	n/a	(qual) QUAN QUAN (qual)	(quan) QUAL QUAL (quan)

Explanatory sequential design

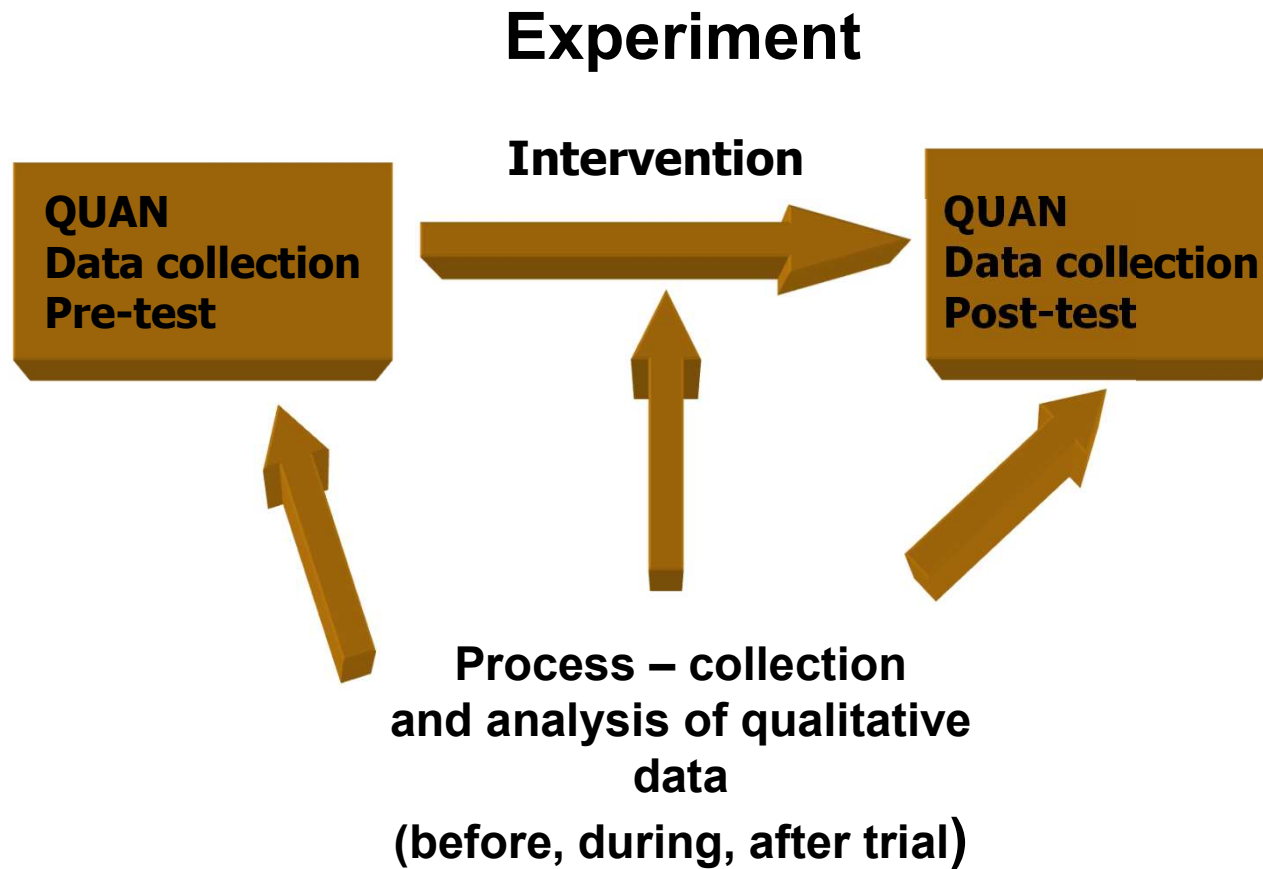


<u>Quantitative Data*</u>	<u>Quantitative Analysis</u>	<u>Case Selection</u>	<u>Qualitative Analysis</u>	<u>Interpretation</u>
Number of cigarettes CESD6	Graphic plot of CES D6 scores over time for each participant	Selected 5 cases maximally varying Identified critical months in which smoking varied	Description of each case Identification of life events occurring during critical months where smoking increased or decreased Thematic analysis of life events for each case Cross-case thematic analysis	Why did changes in smoking occur?
<u>Qualitative Data*</u> Semistructured interviews, audio recorded and transcribed	Graphic plot of cigarettes/day values over time for each participant			

* Data collected 10 times over the course of a calendar year for 40 participants

Source: Creswell, Plano Clark, Shope, McVea. (in progress)

Embedded research design



Mixing the quan and qual data

Type of Mixing	Type of Design	Why Mixing Occurs	Where Mixing Occurs in Research Process
Connecting	Sequential	One phase builds on the other	Between data analysis (Phase 1) and data collection (Phase 2)
Merging	Concurrent	Bring results together	After analysis of both quan and qual – typically in discussion
Embedding	Sequential or Concurrent	Either building or bringing results together	Either between phases or in discussion after analysis

Write a purpose statement for mixed method

This mixed methods study will address _____ (overall content aim of the study). An embedded mixed method design will be used, and it is a design in which one data set provides a supportive, secondary role in a study based primarily on the other data set. The primary purpose of this study will use _____ (quantitative instruments) to test the theory of _____ (the theory) that predicts that _____ (independent variables) will influence _____ (positively, negatively) the _____ (dependent variables or outcomes) for _____ (participants) at _____ (the research site). A secondary purpose will be to gather qualitative data _____ (type of qualitative data, such as interviews) that will explore _____ (the central phenomenon) for _____ (participants) at _____ (site). The reason for collecting the secondary database is _____ (e.g., to address different question, to provide support for the primary purpose).

Writing research questions/hypotheses in mixed methods research

- ◊ Write qualitative research questions and write quantitative research questions/hypotheses
- ◊ Also write a mixed methods research question
- ◊ Write these questions separately

A mixed methods question

Three ways to write this question:

Methodologically-focused:

- ◊ To what extent do the qualitative results confirm the quantitative results?

Content-focused:

How do the interviews with adolescent boys support the quantitative results that their self-esteem changes during the middle school years?

Hybrid of quantitative and qualitative elements:

- ◊ What results emerge from comparing the exploratory qualitative data about boy's self-esteem with outcome quantitative instrument data measured on a self-esteem instrument?

THANK YOU

Prepared by: HSM, NAG and SUH